

Time-of-day immunochemotherapy in nonsmall cell lung cancer: a randomized phase 3 trial

In the format provided by the
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Contents for Supplementary Information

1. Study Protocol (SOP) and Statistical Analysis Plan (SAP)

Effect of Time-of-Day (ToD) for Immunotherapy on Progression-Free Survival in NSCLC

Phase: Phase 3

Clinical Study Leading Site: Hunan cancer hospital

Principal Investigator: Professor Yongchang Zhang

Version No: Version 1.0

Version Date: January 2, 2022

Sponsor: Hunan cancer hospital

VERSION HISTORY/AMENDMENT HISTORY

Document	Version Date	Rationale for Amendment and Summary of Changes
V1.0	January 2, 2022	

Protocol Signature Page of Principal Investigator

I will conscientiously perform my responsibilities as an investigator in accordance with the ICH-GCP regulations and personally participate in or direct this clinical study. I have received the Investigator's Brochure for the study drug in this clinical trial; I have known and read about the preclinical studies of the investigational product and the protocol of this clinical trial. I agree to perform my duties in accordance with ICH-GCP, the Declaration of Helsinki, any applicable laws and regulations, and this protocol. Unless measures must be taken to protect the safety, rights and interests of the subjects, I will only make amendments to the protocol after informing the sponsor and obtaining consent, and the amendments will only be implemented upon approval by the Ethics Committee. I will be responsible for making clinically relevant medical decisions to ensure that the subjects receive appropriate treatment promptly when adverse events occur during the study, and recording and reporting these adverse events in accordance with relevant national regulations. I undertake to record the data in a truthful, accurate, complete and timely manner. I will accept the monitoring and auditing by the monitors or auditors dispatched by the sponsor and the inspection by drug regulatory authorities to ensure the quality of the clinical trial. I undertake to keep the subjects' personal information and related matters confidential. I agree to disclose my full name and occupation to the sponsor, agree to disclose expenses related to the clinical study upon request, and agree to prohibit commercial and economic activities related to this study. I will provide the principal investigator's curriculum vitae before the start of the study, which will be submitted to the Ethics Committee and also the drug regulatory authority for filing.

Study Site: Hunan cancer Hospital

Yongchang Zhang

Principal Investigator (Print Name)

Principal Investigator (Signature)

Signature Date

(MM/DD/YYYY)

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PROTOCOL SYNOPSIS

Study Title	Effect of Time-of-Day (ToD) for Immunotherapy on PFS in NSCLC
Protocol Number	ToD-001
Version Number	V1.0
Sponsor	Hunan cancer hospital
Study Leading Site	Hunan cancer hospital
Principal Investigator	Professor Yongchang Zhang
Participating Sites	1
Study Objectives	Primary objective: - To evaluate and compare the progression-free survival (PFS) of patients with untreated stage IIIC-IV non-small cell lung cancer (NSCLC) receiving immunochemotherapy at different infusion times within a day. PFS will be defined as the interval from randomization to the first documented instance of disease progression or death from any cause, as assessed by a blinded independent review committee (BIRC). Secondary objectives: - To assess and compare the overall survival (OS) of patients with untreated stage IIIC-IV non-small cell lung cancer (NSCLC) receiving immunochemotherapy at different infusion times within a day, define as randomization to the death of the subject due to any cause. - To assess and compare the overall response rate of immunochemotherapy in patients with untreated stage IIIC-IV NSCLC receiving treatment at different infusion times within a day, as assessed by BIRC. Efficacy will be evaluated by the BIRC according to the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, and will be defined as the proportion of subjects achieving a complete response (CR) or a partial response (PR). Exploratory objectives: - To investigate the association between changes in peripheral blood lymphocyte subsets during the treatment and infusion time.
Study Endpoints	Primary study endpoints: - Progression-free survival (PFS) of patients with untreated stage IIIC-IV NSCLC receiving immunochemotherapy at different infusion times during the day. PFS will be defined as the interval from randomization to: ➤ Tumor progression determined by the BIRC per RECIST v1.1 ➤ Death due to any cause Secondary efficacy endpoints

	- Overall survival (OS) of patients with untreated stage IIIC-IV NSCLC receiving immunotherapy at different infusion times during the day. - Objective response rate (ORR) of in patients with untreated stage IIIC-IV NSCLC receiving immunotherapy at different infusion times during the day. Exploratory endpoints: - The relationship between changes in peripheral blood lymphocyte subsets during treatment, infusion time, and treatment efficacy.
Study Subjects	Patients with newly diagnosed stage IIIC-IV non-small cell lung cancer who are treatment-naïve.
Study Design	This is a Phase 3, single-center randomized clinical trial designed to evaluate the efficacy of early daily infusion versus late daily infusion of immunochemotherapy in the treatment of stage IIIC-IV non-small cell lung cancer. The study includes a screening period (from the time the subject signs the informed consent up to 28 days before the first dose), a treatment period (comprising the first 4 cycles of chemotherapy combined with immunotherapy, followed by a maintenance phase after 4 cycles), and a follow-up period (focused on survival). The study schematic is as follows: <pre> graph LR A["1. Time: 2022-2023 2. Number of patients: 210 NSCLC 3. Stage: IIIC-IV 4. ICIs: pembrolizumab or sintilimab"] -- "Random 1:1" --> B["Early ToD group (n=105) Initial 4 infusions before 15:00"] A -- "Random 1:1" --> C["Late ToD group (n=105) Initial 4 infusions after 15:00"] B --> D["Primary endpoint PFS secondary endpoint OS ORR"] C --> D </pre> Screening Period Subjects will be screened and evaluated within 28 days prior to the first dose to determine whether they meet the study criteria. Treatment Period Upon enrollment into this study, subjects will receive treatment in 2 period in sequence: combination period and maintenance period <u>Combination period</u> Upon enrollment into this study, subjects will be randomized according to our trial protocol. They will be allocated in a 1:1 ratio to receive either early ToD infusion or late ToD infusion. All patients will receive 4 cycles of chemotherapy in combination with pembrolizumab or sintilimab. Subjects randomized to the early ToD group will complete their immunotherapy by 15:00, while those in the late ToD group will receive their immunotherapy after 15:00. Chemotherapy will be administered after the immunotherapy infusion. All enrolled patients received 200 mg of pembrolizumab or sintilimab in 21-day cycles. For adenocarcinoma and squamous cell carcinoma, pemetrexed or nab-paclitaxel was used alongside carboplatin and immunotherapy as per guidelines. Specifically: Squamous cell carcinoma: Patients were treated with carboplatin (AUC 5 mg/mL/min) and nab-paclitaxel (200 mg/m²) on day 1, in parallel with immunotherapy. Adenocarcinoma: Patients received carboplatin (AUC 5 mg/mL/min) and pemetrexed (500 mg/m²) on day 1, along with immunotherapy. <u>Maintenance period</u>

	After 4 cycles of treatment, patients will enter maintenance therapy if no disease progression occurs. For patients with histologically confirmed squamous cell carcinoma of the lung, pembrolizumab or sintilimab monotherapy will be administered in 21-day cycles, continuing until disease progression or intolerable adverse events. For patients with histologically confirmed adenocarcinoma of the lung, maintenance therapy will consist of pemetrexed in combination with pembrolizumab or sintilimab, also administered in 21-day cycles, until disease progression or intolerable adverse events. During the maintenance phase, the timing of immunotherapy infusion will not be strictly fixed. Follow-up Period After completing treatment, subjects will enter the follow-up period, which will primarily focus on survival. Telephone follow-ups will be conducted every 3 months to gather information on the subjects' survival status, any anti-tumor treatments received after the study, time to disease progression, and other relevant data. This follow-up will continue until the subject's death, loss to follow-up, withdrawal of informed consent, or termination of the study by the sponsor. Tumor Imaging Assessment Tumor assessments will be conducted periodically based on RECIST v1.1 criteria. All subjects will undergo initial tumor imaging assessments during screening using PET-CT (preferred) or enhanced neck/chest/abdomen CT and enhanced head MRI. Once enrolled in the study, patients will be evaluated after every two treatment cycles, with imaging including enhanced CT scans of the chest and abdomen, as well as head MRI. During the maintenance therapy phase, tumor assessments will follow the same schedule as during the chemotherapy and immunotherapy combination phase, continuing until the end of maintenance therapy. During the follow-up period, tumor assessments will not be emphasized; only the patients' survival status will be recorded. However, the frequency of assessments may be appropriately increased in the event of adverse events or possible disease progression, as deemed necessary by the investigator.
Study Drug	immunotherapy drugs • Pembrolizumab injection (“pembrolizumab”) • Sintilimab injection (“sintilimab”) Chemotherapy drugs • Nab-Paclitaxel injection (“nab-paclitaxel”) • Pemetrexed injection (“pemetrexed”)) • Carboplatin injection (“carboplatin”)
Route of Administration	Pembrolizumab/Sintilimab: Administer 200 mg of pembrolizumab or sintilimab via intravenous infusion on Day 1 of each 3-week (21-day) cycle, over approximately 30-60 minutes. Intravenous bolus or rapid infusion is not permitted. At the end of the infusion, flush the infusion line with an adequate volume of physiological saline according to the medical standards of the research center. Adjust the administration time based on patient tolerance and immune-related adverse events. Treatment should be discontinued if the patient cannot tolerate the drug or experiences Grade 4 or higher adverse events. Chemotherapy Regimen: nab-paclitaxel Injection: Administer 200 mg/m² of nab-paclitaxel via intravenous

	infusion on Day 1 of each 3-week (21-day) cycle. Dose adjustments and delays are permitted. Infusion may be delayed for up to 9 weeks from the last dose; otherwise, treatment should be discontinued. Pemetrexed Injection: Administer 500 mg/m² of pemetrexed via intravenous infusion on Day 1 of each 3-week (21-day) cycle. Dose adjustments and delays are permitted. Infusion may be delayed for up to 9 weeks from the last dose; otherwise, treatment should be discontinued. Carboplatin Injection: Administer carboplatin at AUC 5 mg/(mL·min) via intravenous infusion on Day 1 of each 3-week (21-day) cycle. Dose adjustments and delays are permitted. Infusion may be delayed for up to 9 weeks from the last dose; otherwise, treatment should be discontinued. Combination Therapy Administration: During combination therapy with pembrolizumab/sintilimab and chemotherapy, the order of administration on Day 1 of each 3-week (21-day) cycle will be as follows: Pembrolizumab/Sintilimab → nab-Paclitaxel/Pemetrexed → Carboplatin Ensure there is a minimum interval of 30 minutes between the administration of pembrolizumab/sintilimab and nab-paclitaxel/pemetrexed.
Peripheral Blood Collection and test	For each patient participating in the study, a complete blood count will be performed at baseline and before each treatment cycle to assess hematological adverse events caused by the drug and to determine whether any necessary interventions are required before proceeding to the next treatment cycle. Additionally, peripheral blood lymphocyte subset analysis will be conducted with the patient's informed consent and voluntary participation. These assessments will be performed at baseline, after 2 cycles of treatment (before 3rd cycle treatment), and after 4 cycles of treatment (before 5th cycle treatment). Blood samples will be collected between 8:00 AM and 10:00 AM on specified days before the next infusion. Flow cytometry will be used to determine the relative frequencies of specific lymphocyte subsets within the total peripheral blood lymphocytes. These subsets include total T cells (CD3+), CD4+ T cells (CD3+, CD4+), CD8+ T cells (CD3+, CD8+), NK cells (CD16+ CD56+), and B cells (CD20+).
Inclusion Criteria	Patients must meet all of the following inclusion criteria to be eligible for entry into the study. 1. Understanding the requirements and content of the clinical trial, and providing a signed and dated informed consent form. 2. Male or female patients aged ≥18 years at the time of signing the informed consent form. 3. An Eastern Cooperative Oncology Group (ECOG) performance status of 0-1. 4. Histologically or cytologically confirmed stage IIIC-IV non-small cell lung cancer (NSCLC) with no prior systemic treatment. Disease staging should be based on the 8th edition of the AJCC/UICC NSCLC staging system. 5. Presence of measurable target lesions as defined by RECIST 1.1, confirmed by CT or MRI within 28 days prior to the first dose of study drug. 6. Negative for EGFR, ALK, and ROS1 sensitizing alterations, as confirmed by a certified laboratory. 7. Patients with stable brain metastases are eligible if no local treatment or steroid is required. 8. FEV1 > 1.0 L and FEV1% > 40%. 9. Adequate hematologic and major organ function defined by the following laboratory criteria, with tests completed within 14 days prior to the first dose of study

	treatment: (1) Hematology (no hematopoietic stimulating factor drugs or blood transfusion within 14 days before the first study treatment): absolute neutrophil count (ANC) $\geq 1.5 \times 10^9/L$, absolute lymphocyte count (LC) $\geq 0.5 \times 10^9/L$; platelet count (PLT) $\geq 100 \times 10^9/L$, hemoglobin (Hb) ≥ 90 g/L (2) Liver function: aspartate transferase (AST) and alanine aminotransferase (ALT) $\leq 3 \times$ ULN; serum total bilirubin (TBIL) $\leq 1.5 \times$ ULN (total bilirubin ≤ 3.0 mg/dL in patients with confirmed Gilbert syndrome); albumin (ALB) ≥ 3 g/dL; (3) Renal function: creatinine clearance rate (CrCl) ≥ 45 mL/minute (by Cockcroft-Gault formula); (4) Coagulation: international normalized ratio (INR) ≤ 1.5, activated partial thromboplastin time (APTT) $\leq 1.5 \times$ ULN; (5) Cardiac color ultrasound: left ventricular ejection fraction (LVEF) $\geq 50\%$ 10. Female or male patients of childbearing potential who are not surgically sterilized must agree to abstain from sexual intercourse (avoid heterosexual intercourse) or use at least one medically acceptable contraceptive (e.g., bilateral tubal ligation, male sterilization, hormonal contraceptives that inhibit ovulation, intrauterine device that releases hormones and copper-containing intrauterine device) during the study treatment period and for 180 days after the end of the study treatment period; female patients of childbearing potential who are not surgically sterilized must have a negative serum HCG test within 14 days before the first dose.
Exclusion Criteria	Patients with any of the following characteristics or conditions will not be eligible for the study: Tumor-Related and Anti-Tumor Treatment Exclusions: 1. Patients with histologically or cytologically confirmed small cell lung cancer (SCLC), mixed SCLC and NSCLC, large-cell neuroendocrine carcinoma, or sarcomatoid carcinoma. 2. Patients with EGFR mutations or ALK/ROS1 gene rearrangements. All patients must undergo EGFR gene testing and ALK/ROS1 gene and/or immunohistochemistry testing. 3. Patients who have previously received systemic anti-tumor therapy for NSCLC, including investigational drugs. Additionally, those who have received traditional Chinese medicine for anti-tumor purposes must ensure a washout period of at least 2 weeks prior to the first dose of the study drug. 4. Patients who have previously received radiotherapy to the chest. 5. Patients who have participated in other clinical trials within 4 weeks or within 5 drug half-lives (whichever is shorter) prior to the first dose of the study drug. This includes those who received systemic immunostimulatory therapy (e.g., interferon or interleukin-2, including investigational immunostimulatory agents) within 4 weeks prior to the first dose. 6. Patients who received systemic immunosuppressive therapy (e.g., corticosteroids, cyclophosphamide, azathioprine, methotrexate, thalidomide, anti-tumor necrosis factor drugs) within 2 weeks prior to the first dose or who are expected to require such therapy during the study. Exceptions may be made for patients who received short-term, low-dose systemic immunosuppressive drugs (≤ 10 mg/day prednisone or equivalent) or short-term, high-dose systemic corticosteroids (e.g., for contrast agent allergies, 48-hour corticosteroid treatment), with the approval of the medical monitor. Patients with chronic obstructive pulmonary disease (COPD) on ≤ 10 mg/day prednisone or equivalent, patients with orthostatic hypotension on mineralocorticoid therapy (e.g., fludrocortisone), and patients with adrenal

	insufficiency on physiological replacement doses of corticosteroids may be enrolled. 7. Patients with symptomatic brain metastases requiring dehydration therapy or local radiotherapy. Comorbidities/medical history 8. Presence of an autoimmune disorder. The following are allowed: Type 1 diabetes (blood glucose can be controlled by insulin therapy) and hypothyroidism caused by autoimmune thyroiditis for which only hormone replacement therapy is indicated; vitiligo skin disease manifestations only (psoriatic arthritis should be excluded); 9. Presence of a malignant tumor other than non-small cell lung cancer within 5 years before screening, with the exception of adequately treated cervical carcinoma in situ, basal cell or squamous epithelial cell skin cancer, local prostate cancer after radical surgery, and ductal carcinoma in situ after radical surgery (hormone treatment of non-metastatic prostate cancer or breast cancer is allowed); 10. Patients with known or suspected interstitial pneumonitis; other moderate to severe lung diseases that may interfere with the detection or treatment of drug-related pulmonary toxicity and that seriously affect respiratory function, including, for example, idiopathic pulmonary fibrosis, organizing pneumonia/bronchiolitis obliterans, etc.; 11. Patients with severe cardiovascular and cerebrovascular diseases, such as conditions meeting NYHA criteria (Class II or higher), or myocardial infarction or cerebrovascular accidents (cerebral hemorrhage, symptomatic cerebral infarction, etc.) within 3 months before the first dose, or unstable arrhythmia or unstable angina within 1 month before the start of study treatment; 12. Clinically significant bleeding symptoms or a significant bleeding tendency within one month before the first dose, e.g., gastrointestinal bleeding, gastric ulcer bleeding, etc.; 13. Arterial/venous thrombosis event within 3 months before the first dose, such as deep vein thrombosis and pulmonary embolism; 14. Positive HIV test; 15. Patients with active hepatitis B (defined as positive hepatitis B virus surface antigen [HBsAg] during the screening period along with HBV DNA above the upper limit of normal of the site's laboratory) or hepatitis C (defined as positive hepatitis C virus antibody [HCV-Ab] and positive HCV-RNA during the screening period); 16. Evidence of active tuberculosis infection within 1 year before the first dose; 17. Serious infection within 4 weeks before the first dose, including but not limited to infection complications, bacteremia and severe pneumonia requiring hospitalization or ≥ 2 weeks of intravenous antibiotic administration; active infection that has received therapeutic intravenous or oral antibiotics within 2 weeks before the start of study treatment. Patients receiving prophylactic antibiotic therapy (e.g., to prevent urinary tract infection) may be enrolled 18. History of live attenuated vaccination within 28 days before the first dose or patients expecting live attenuated vaccination during the study; 19. Major surgery other than diagnosis or biopsy within 28 days before the first dose; minor traumatic surgery (biopsy, bronchoscopy and chest drainage) within 7 days before the first dose; 20. Patients who have received or are preparing to receive an allogeneic bone marrow transplantation or solid organ transplantation; 21. History of severe allergic reactions to other monoclonal antibodies/fusion protein drugs; 22. Hypersensitivity to any component of the randomly assigned treatment regimen;
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	23. Female patients who are pregnant, breastfeeding, or planning to become pregnant during the study; 24. Patients with a known history of psychotropic substance abuse, alcohol or drug use; and any other medical (e.g., pulmonary, metabolic, endocrine or neurological disease, congenital disease, etc.), psychiatric, or social condition that, in the opinion of the investigator, may interfere with the patient's rights, safety, health, or ability to sign the informed consent form or collaborate and participate in the study, or interfere with the evaluation of the study drug, or the interpretation of patient safety or study results.
Study Withdrawal Criteria/ Termination of Study Treatment Criteria	Subjects must withdraw from the study in the following cases: • The subject or their legal representative withdraws informed consent to participate in the study. • The subject is lost to follow-up. Subjects must discontinue the study drug but continue follow-up for the following reasons: • Intolerable adverse reactions related to the study drug. • Tumor progression confirmed by imaging or clinical symptoms suggestive of progression. • General deterioration of health status, making continued participation in the study unfeasible. • Occurrence of pregnancy in female subjects during the study. • The subject initiates other systemic anti-tumor treatments. • The investigator determines that the subject is unable to continue treatment for other reasons, such as mental illness, social conditions that might interfere with the subject's rights, safety, cooperation, and participation in the study, or poor compliance.
Sample Size Calculation	The primary endpoint of this study is the Progression-Free Survival (PFS). The sample size is determined based on the PFS of the two groups. The study design includes two cohorts: the Early ToD group, where the first four immunotherapy infusions are administered before 15:00, and the Late ToD group, where the first four immunotherapy infusions are administered after 15:00. Based on data from our previous retrospective studies, it is assumed that the median PFS for the Late ToD group is 6 months, while the median PFS for the Early ToD group is 10 months. The two groups will be randomized in a 1:1 ratio. Statistical assumptions include a two-sided $\alpha = 0.05$ and 80% power. The enrollment period is assumed to be 12 months, with a 24-month follow-up period and a dropout rate of 5%. The final calculated total sample size for the two groups is 210 patients, including 105 patients for Early ToD group and 105 patients for Late ToD group.
Data Analysis and Statistical Methods	Analysis sets: Intent-to-Test (ITT) Population: This population includes all randomized patients, and analyses are performed based on their originally assigned treatment groups, regardless of whether they completed the intervention as planned. This dataset serves as the primary population for evaluating efficacy outcomes. The Full Analysis Set (FAS): FAS included all randomized patients who received at least one dose of the study treatment as planned. Per-protocol Set (PPS): PPS includes patients who strictly adhere to the treatment regimen, excluding subjects with major protocol deviations that are judged to have a significant impact on the results. General analysis: Categorical data will be presented descriptively through statistics such as frequency and percentage, with a 95% confidence interval provided for overall percentages when applicable. Continuous data will be summarized descriptively using statistics including the mean, standard deviation (SD), standard error of the mean (SEM), median, as well as minimum and maximum values. Time-to-event data will be analyzed using the Kaplan-Meier method to estimate the survival function, generate survival curves, and determine the median time along with its

	95% confidence interval. For the statistical analysis, group comparisons will be conducted between the Early and Late ToD groups, unless stated otherwise. Additionally, subgroup analyses will be carried out based on the baseline characteristics of the patients throughout the study. Efficacy analysis: Analysis of first efficacy endpoints: The primary efficacy endpoints of this study are the PFS of untreated stage IIIC-IV NSCLC receiving immunochemotherapy infused during the early ToD and Late ToD. PFS: Survival curves will be generated utilizing the Kaplan-Meier method, and the median survival time will be calculated for each cohort. Standard errors will be determined using Greenwood's formula, while two-sided 95% confidence intervals for the median survival time will be derived for each cohort through the log-log transformation method. Analysis of secondary efficacy endpoints: OS: Survival curves will be created using the Kaplan-Meier method, and the median survival time will be estimated for each cohort. Standard errors will be calculated using Greenwood's formula, and two-sided 95% confidence intervals for the median survival time will be obtained for each cohort through the log-log transformation approach. ORR: Descriptive statistics will summarize the ORR for each cohort. Both the Clopper-Pearson and Wilson methods will be employed to estimate the 95% confidence intervals for the ORR of each cohort. For subjects who do not complete the ORR assessment, the imputation method will be detailed in the statistical analysis plan. Exploratory analysis: The two-way ANOVA and linear regression will be utilized to investigate the relationship between changes in peripheral blood lymphocyte subsets during treatment and infusion time.
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SCHEDULE OF ACTIVITIES

Study Procedure/Visit	Screening Period (Day - 28)					Maintenance Treatment Period		Follow-up Period	
	Day - 28	C 1	C 2	C 3	C 4	MC1	MC2- End of treatment	End of treatment/ withdrawal	Survival Follow-up Every 3 months
Time window						3d		7d	
Administrative procedures									
Signing of informed consent form ^[1]	√								
Demographics ^[2]	√								
Tumor diagnosis ^[3]	√								
Prior medical history	√								
Checking against inclusion and exclusion criteria ^[4]	√								
Blood sample collection ^[5]	√	√				√			
Follow-up of survival information and anti- tumor treatment ^[6]									√
Clinical procedures/assessments									

[illegible]

Hematology ^[13]	√	√	√	√	√	√	√		
Study Procedure/Visit	Screening Period (Day - 28)					Maintenance Treatment Period		Follow-up Period	
	Day - 28	C 1	C 2	C 3	C 4	MC1	MC2- End of treatment	End of treatment/ withdrawal	Survival Follow- up Every 3 months
Biochemistry ^[14]	√	√	√	√	√	√	√		
Urinalysis ^[15]	√	√	√	√	√	√	√		
Fecal occult blood ^[16]	√	√	√	√	√	√	√		
Coagulation ^[17]	√	√	√	√	√	√	√		
Thyroid function test ^[18]		√		√			√		
Efficacy assessment									
Tumor imaging examination ^[19]		√		√			√	√	

Tumor progression follow-up ^[20]		√		√			√	√	√
Study Procedure/Visit	Screening Period (Day - 28)					Maintenance Treatment Period		Follow-up Period	
	Day - 28	C 1	C 2	C 3	C 4	MC1	MC2- End of treatment	End of treatment/ withdrawal ^[24]	Survival Follow-up Every 3 months ^[25]
Study drug administration									
Pembrolizumab/Sintilimab administration ^[21]		√	√	√	√	√	√		
nab-paclitaxel administration ^[22]		√	√	√	√				
Pemetrexed administration ^[23]		√	√	√	√	√	√		
Cisplatin administration ^[24]		√	√	√	√				

Note 1: Repeat testing is not required during the screening period if the tests (e.g., tumor imaging, tumor tissue biopsy, pathological diagnosis, etc.) completed by the subject at the site prior to signing the ICF are within the time frame specified in the study and are in compliance with study regulations.

Note 2: In addition to the test items and time points listed in the table, the investigator may add necessary test items as necessary.

1. Prior written informed consent forms must be obtained before any clinical study procedures are performed. Re-screening is allowed in this study for subjects who have previously failed screening. At re-screening, they must sign the informed consent form again and re-register to obtain a new subject

number.

2. This includes date of birth, age, gender, ethnicity, etc.
3. Histopathological or cytological diagnostic results must be collected, including pathological specimen type and method of acquisition, pathological subtype, histological classification, clinical stage and TNM stage, primary lesion, and time of first pathological diagnosis. For nodal status assessment, if PET-CT revealed suspicious mediastinal nodal involvement, pathological confirmation (via mediastinoscopy, thoracoscopy, or endobronchial ultrasound) was strongly recommended. EGFR gene test results and ALK/ROS1 gene test and/or immunohistochemistry results must be collected for all subjects. For subjects with lung adenocarcinoma, a histological test must be used for EGFR and ALK/ROS1 mutations. For subjects with lung squamous cell carcinoma, a histological test is preferred if the tissues are sufficient. If a liquid biopsy is required, consult with the sponsor.
4. Subjects must be checked against the inclusion and exclusion criteria before the first dose.
5. Blood sample collection: Refer to the Flowchart of PK/Immunogenicity Blood Sample Collection.
6. Follow-up on survival information and anti-tumor treatment*: This includes follow-up on tumor progression. Subjects without evidence of radiographic disease progression at the end of treatment will still continue to undergo imaging assessments according to the frequency of clinical tumor imaging assessment required during the treatment period, until evidence of radiographic disease progression or death of the subject, loss to follow-up, or withdrawal of informed consent.
7. Adverse events: All adverse events that occur after signing of the informed consent form will be observed and recorded as per NCI CTCAE v5.0 until 90 ± 7 days after the last dose immunotherapy. Refer to the table 1. Principles of Collection and Follow-up Periods for AE/SAE/SIE/Pregnancy Events, for the specific AE collection periods. All adverse events must be followed up until recovery or return to baseline level, or stable disease, or $\leq$ grade 1 (CTCAE v5.0), or until the subject is lost to follow-up or dies, or the event is finally confirmed to be unrelated to the study drug or study process at the end of the safety follow-up period. Every effort should be made to ensure that the subjects get the best outcome and a clear causal relationship assessment.
8. Physical examination and body weight: A general physical examination will be performed during the screening period, including: general condition, head, face, eyes, ears, nose, throat, oral cavity, skin, lymph nodes, respiratory system, cardiovascular system, gastrointestinal system, genitourinary system, musculoskeletal system, nervous system, psychiatric state and others. Abnormal findings observed at baseline (within 14 days prior to the first dose) will be recorded as part of the general medical history and baseline disease in the eCRF. A directed physical examination may be performed at subsequent visits, and clinically significant new or worsening abnormal results will be recorded in the adverse event form of the eCRF. A physical examination must be performed within 3 days before the first dose, within 14 days before local treatment, and within 3 days before each dose. Body weight must be recorded within 3 days before the first dose, within 14 days before local treatment, and within 3 days before each dose.
9. Vital signs should include temperature, pulse, respiratory rate, and blood pressure. During the treatment period, a vital signs examination is required within

3 days before the first dose, within 14 days before local treatment, and within 30 min before ICIs infusion and within 30 - 60 min after all drugs have been infused on the day of dosing in each cycle. During cisplatin infusion, the clinical practice should be followed, but the result is not required to be recorded in the eCRF.

10. Oxygen saturation: During the treatment period, it should be measured within 3 days before each dose, within 14 days before local treatment, and when the subject experiences any new respiratory symptoms or worsening of respiratory symptoms. Oxygen saturation will be determined by resting pulse measurement.
11. ECOG: During the treatment period, it should be scored within 3 days before the first dose, within 14 days before local treatment, and within the first 3 days of each cycle.
12. 12-lead ECG: It should be performed within 14 days before the first dose. Pay attention to the heart rate and QT and P-R intervals. If chest pain, palpitations or other heart symptoms occur, an additional examination can be performed at any time. A 12-lead ECG examination must be performed within 3 days before dosing in each cycle (except for the first dose) and within 14 days before local treatment during the treatment period.
13. Hematology: It includes complete and differential blood counts. It should be tested within 14 days before the first dose (screening period), within 3 days before the first dose (no need for a repeat test if the hematology test during the screening period is completed within 3 days before the first dose), within 14 days before local treatment, and within 3 days before dosing in each cycle during the treatment period.
14. Biochemistry: It includes ALT, AST, ALP, LDH, TP, ALB, TBIL, DBIL, GLU, BUN or urea, UA, Cr, CK, K⁺, Na⁺, Cl⁻, Ca²⁺, and Mg²⁺. It will be performed within 14 days before the first dose. During the treatment period, the test will be performed within 3 days before dosing in each cycle (no need for a repeat test if the biochemistry test during the screening period is completed within 3 days before the first dose) and within 14 days before local treatment.
15. Urinalysis: It includes urine pH, specific gravity, white blood cells, red blood cells, and urine protein. It will be performed within 14 days before the first dose. If urine protein is $\geq 2+$, an additional 24-hour urine protein quantification test is required, and enrollment of subjects with 24-hour urine protein quantification ≥ 1 g is not recommended. During the treatment period, the test will be performed within 3 days before dosing in each cycle (except for the first dose) and within 14 days before local treatment.
16. Fecal occult blood: It will be performed within 14 days before the first dose. If positive, a re-test must be performed. If the re-test result is fecal occult blood +, the investigator should perform gastroscopy as clinically necessary to rule out gastrointestinal bleeding.
17. Coagulation: It includes international normalized ratio (INR), activated partial thromboplastin time (APTT), fibrinogen (FIB), and thrombin time (TT). If INR is not available, prothrombin time (PT) can be used instead. It will be performed within 14 days before the first dose. During the treatment period,

the test will be performed within 3 days before dosing in each cycle (except for the first dose) and within 14 days before local treatment.

18. Thyroid function test: It includes serum thyroid-stimulating hormone (TSH), free triiodothyronine (FT3) and free thyroxine (FT4). If FT3 and FT4 are not available, triiodothyronine (T3) and thyroxine (T4) can be used instead. It will be performed within 14 days before the first dose. The test will be performed within 3 days before dosing every 2 cycles during the treatment period, and will be performed from the MC1 cycle during the maintenance treatment period.
19. Tumor imaging assessment: Tumor imaging evaluation: Tumor imaging evaluation will be performed according to RECISTv1.1 during the screening period and at regular intervals after treatment. Subjects will undergo a complete tumor imaging assessment during screening using PET-CT and enhanced head MRI. After entering the study, imaging assessment will be performed by chest and abdomen CT and head MRI every 2 cycles after anti-tumor treatment. If a subject experiences further progression on the next tumor assessment and the subject is no longer benefiting from continued treatment, the subject should be withdrawn from study treatment. The frequency of assessments may be increased according to the investigator's plan if clinically indicated. Subjects should be evaluated using the same imaging techniques throughout the treatment period and all imaging data should be retained.
20. Pembrolizumab/Sintilimab administration: Pembrolizumab/Sintilimab was administered at a fixed dose of 200 mg every cycle, including the combination chemotherapy treatment period as well as the maintenance treatment period, until disease progression or serious adverse events occurred.
21. Nab-paclitaxel administration: nab-paclitaxel 200 mg/m² was administered intravenously on Day 1 of each 3-week dosing cycle. The drug was administered for 4 consecutive cycles, and the drug was discontinued if disease progression or serious adverse events occurred during this period.
22. Pemetrexed administration: Pemetrexed was administered at a dose of 500 mg/m² every 3 weeks in combination with carboplatin and immunotherapy for 4 cycles, followed by maintenance therapy with pemetrexed and immunotherapy until disease progression or serious adverse events.
23. Carboplatin administration: Carboplatin AUC 5 is administered via intravenous drip on Day 1 of each 3-week dosing cycle. The drug was administered for 4 consecutive cycles, and the drug was discontinued if disease progression or serious adverse events occurred during this period.
24. End of treatment/withdrawal: Visits will be completed within ± 7 days of confirmed end of treatment/withdrawal.
25. Survival follow-up will be performed over phone every 3 months (± 14 days) after the last dose (regardless of the study drug). Information on the survival (date of death and cause of death) of subjects and after the end of study treatment (including subsequent anti-tumor treatment and time to progression) will be collected until the death endpoint or until the subject is lost to follow-up or the study is terminated.

ABBREVIATIONS

Abbreviations	Full Name in English	Full Name in Chinese
ACTH	Adrenocorticotrophic hormone	促肾上腺皮质激素
AE	Adverse event	不良事件
ALB	Albumin	白蛋白
ALP	Alkaline phosphatase	碱性磷酸酶
ALT	Alanine aminotransferase	谷氨酸丙氨酸氨基转移酶
ANC	Absolute neutrophil count	中性粒细胞计数
APTT	Activated partial thromboplastin time	活化部分凝血活酶时间
AST	Aspartate aminotransferase	谷氨酸天门冬氨酸氨基转移酶
AUC	Area under curve	曲线下面积
BUN	Blood urea nitrogen	尿素氮
Ca ²⁺	Serum calcium	血钙
CK	Creatine kinase	肌酸激酶
Cl ⁻	Blood chlorine	血氯
CNS	Central nervous system	中枢神经系统
Cr	Creatinine	肌酐
CrCl	Creatinine clearance	肌酐清除率
CRF	Case report form	病例报告表
CTCAE	Common terminology criteria for adverse events	不良事件通用术语标准
DBIL	Direct bilirubin	直接胆红素
DCR	Disease control rate	疾病控制率
EC	Ethics committee	伦理委员会
ECG	Electrocardiogram	心电图
ECOG	Eastern cooperative oncology group	美国东部肿瘤协作组
EO	Eosinophil	嗜酸性粒细胞
FFPE	Formalin fixed paraffin embedded	石蜡包埋组织
FIB	Fibrinogen	纤维蛋白原
FT3	Free triiodothyronine	游离三碘甲状腺原氨酸
FT4	Free thyroxine	游离甲状腺素
GCP	Good Clinical Practice	药物临床试验质量管理规范
GLU	Blood glucose	血糖
Hb	Hemoglobin	血红蛋白
HBcAb	Hepatitis B core antibody	乙肝核心抗体
HBeAb	Hepatitis B e-antibody	乙肝 e 抗体
HBeAg	Hepatitis B e-antigen	乙肝 e 抗原
HBsAb	Hepatitis B surface antibody	乙肝表面抗体
HBsAg	Hepatitis B surface antigen	乙肝表面抗原
HBV	Hepatitis B	乙肝
HCV	Hepatitis C	丙肝
HIV	Human immunodeficiency virus	人类免疫缺陷病毒
HR	Hazard ratio	风险比
ICF	Informed consent form	知情同意书
ICI	Immune checkpoint inhibitor	免疫检查点抑制剂
IDMC	Independent data monitoring committee	独立数据监查委员会
IEC	Independent ethical committee	独立伦理委员会
INR	International normalized ratio	国际标准化比率
irAE	Immune-related adverse event	免疫相关不良事件
IRB	Institutional review board	机构审查委员会
BIRC	Independent review committee	独立影像评估委员会
K ⁺	Serum potassium	血钾
LC	Lymphocyte count	淋巴细胞计数绝对值

Abbreviations	Full Name in English	Full Name in Chinese
LDH	Lactate dehydrogenase	乳酸脱氢酶
LEU	Leukocytes in urine	尿白细胞
LVEF	Left ventricular ejection fraction	左室射血分数
MRI	Magnetic Resonance Imaging	核磁共振成像
Mg ²	Serum magnesium	血镁
Na ⁺	Plasma sodium	血钠
NMPA	National Medical Products Administration	国家药品监督管理局
NSCLC	Non-small cell lung cancer	非小细胞肺癌
OB	Occult blood	大便潜血
ORR	Objective response rate	客观缓解率
OS	Overall survival	总生存期
PFS	Progression-free survival	无进展生存期
pH	Pondus Hydrogenii	酸碱度
PLT	Blood platelet	血小板
PR	Partial response	部分缓解
PRO	Protein in urine	尿蛋白
PT	Prothrombin time	凝血酶原时间
RECIST	Response evaluation criteria in solid tumors	实体肿瘤疗效评价标准
SAE	Serious adverse event	严重不良事件
SAP	Statistical analysis plan	统计分析计划
SG	Specific gravity of urine	尿比重
SIE	Special-interest event	特别关注的不良事件
SOP	Standard operating procedure	标准操作规程
sUA	Serum uric acid	血尿酸
TBIL	Total bilirubin	总胆红素
TP	Total protein	总蛋白
TSH	Thyroid stimulating hormone	血清促甲状腺激素
TT	Thrombin time	凝血酶时间
TTP	Time to progression	至疾病进展时间
UA	Uric acid	尿酸
ULN	Upper limit of normal	正常值上限
URBC	Urine red blood cell	尿红细胞

1 INTRODUCTION: STUDY BACKGROUND AND SCIENTIFIC BASIS

1.1 Study Background

Immune checkpoint inhibitors (ICIs) have significantly improved treatment outcomes in patients with non-small cell lung cancer (NSCLC).^{1,2} However, there remains a subset of patients who do not respond to immunotherapy and will become resistant to immunotherapy.³ Enhancing the effectiveness of immunotherapy remains a crucial focus of current research. Currently, there are several explorations aimed at improving the efficacy of immune checkpoints. While numerous approaches are being investigated to enhance the efficacy of ICIs, including thorough analyses of the immune microenvironment for optimal combination strategies and the identification of predictive biomarkers, these efforts have thus far demonstrated limited effectiveness. Therefore, further exploration is essential to discover more effective methods for improving ICI efficacy.

1.1.1 Circadian rhythm in cancer treatment

Circadian rhythm, which significantly affects the distribution and function of immune cells, plays a role in regulating immune response and may consequently impact the efficacy of immunotherapy.⁴⁻⁸ In recent years, multiple retrospective studies have demonstrated that the efficacy of immunotherapy infusion in the morning is superior to that in the afternoon.⁹⁻¹⁴ A recent meta-analysis of 13 studies has revealed that both overall survival (OS) and progression-free survival (PFS) nearly doubled in patients receiving immune checkpoint inhibitors primarily in the morning or early afternoon, compared to those with later dosing times¹⁵. Remarkably, these time-of-day effects were consistent across seven different tumor types, including non-small cell lung cancer, malignant melanoma, kidney cancer, bladder cancer, esophageal cancer, stomach cancer, and hepatocarcinoma.^{15,16}

1.2 Scientific Rationale for this Study

Numerous retrospective studies involving over two thousand patients have reported puzzling differences in survival and progression-free survival based on the time-of-day administration of immune checkpoint inhibitors^{9,14,15}. A recent meta-analysis of 13 reported studies revealed that both overall survival and progression-free survival were nearly doubled in patients receiving immune checkpoint inhibitors mostly in the morning or early afternoon, compared to later dosing times.¹⁵ Notably, these time-of-day effects were remarkably consistent across patients with seven different tumor types, including non-small cell lung cancer, malignant melanoma, kidney cancer, bladder cancer, esophageal cancer, stomach cancer, and hepatocarcinoma.

In previous retrospective studies, the cutoff values for infusion time were not uniform, ranging from 11:00 to 16:30. The definition of 'before' and 'after' groups also varied across studies. In some cases, 20% to 50% of all infusion times were selected after the cutoff^{14,17}, while in others, 50% of the first three months of infusion times¹⁸ or 25% of the first four infusion times were selected.^{19,20} Despite these inconsistencies, the analyses ultimately yielded consistent results.

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2 STUDY OBJECTIVES AND STUDY ENDPOINTS

2.1 Study Objectives

2.1.1 Primary Objectives

- To evaluate and compare PFS in patients with NSCLC receiving immunochemotherapy infusions administered at early versus later times of day, as assessed by a BIRC.

2.1.2 Secondary Objectives

- To evaluate and compare OS in patients with NSCLC receiving immunochemotherapy infusions at early versus later times of day.
- To evaluate and compare ORR in patients with NSCLC receiving immunochemotherapy infusions at early versus later times of day, as assessed by BIRC.

2.1.3 Exploratory Objectives

- To investigate the relationship between changes in peripheral blood lymphocyte counts during treatment and infusion time.

2.2 Study Endpoints

2.2.1 Primary Study Endpoints

- PFS assessed by the BIRC after immunochemotherapy infused in early ToD or Late ToD, define as randomization to first documented disease progression assessed by BIRC or death whichever occurs first:
 - Tumor progression determined by the BIRC per RECIST v1.1.
 - Death due to any cause.

2.2.2 Secondary Study Endpoints

- OS after immunotherapy infused in early ToD or Late ToD define as randomization to death.
- ORR assessed by the BIRC of immunotherapy infusion in morning infusion and afternoon infusion.

2.2.3 Exploratory Endpoints

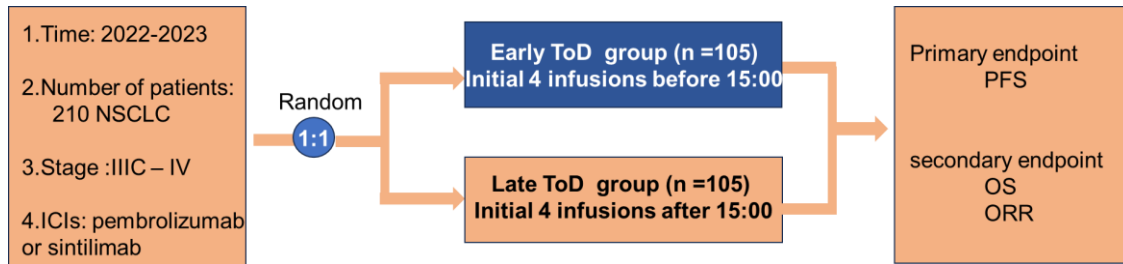
- ORR assessed by the BIRC of immunotherapy infusion during early and later times of the day.
- Relationship between peripheral blood lymphocyte changes during the treatment and the infusion time.

3 STUDY DESIGN

3.1 Overall Design

This is a Phase 3, single-center randomized clinical trial designed to evaluate the efficacy of immunotherapy administered either early or late in the day for patients with Stage IIIC-IV non-small cell lung cancer. The study consists of three phases: a screening period (up to 28 days prior to the first dose, following the subject's signing of informed consent), a treatment period (including both combination therapy and maintenance therapy phases), and a follow-up period (primarily for survival follow-up).

The study schematic is as follows:



Screening Period

Subjects will be screened and evaluated within 28 days prior to the first dose to determine whether they meet the study criteria.

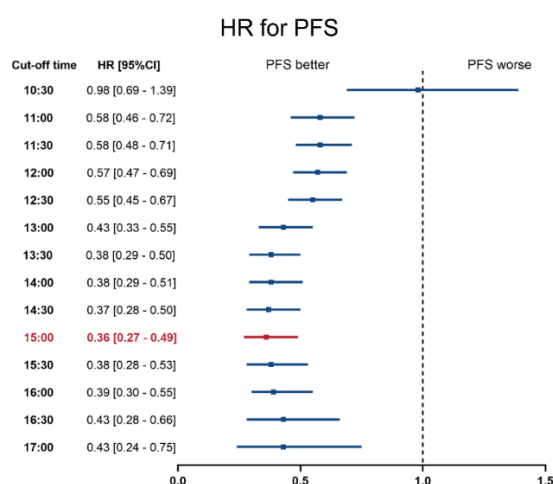
Treatment Period

Upon enrollment into this study, subjects will receive treatment in 2 period in sequence: combination period and maintenance period

Combination period

Upon enrollment into this study, subjects will be randomized according to our trial protocol. They will be allocated in a 1:1 ratio to receive either early infusion (Early ToD group) or late infusion (Late ToD group). All patients will receive 4 cycles of chemotherapy in combination with pembrolizumab or sintilimab. Subjects randomized to the Early ToD group will complete their immunotherapy by 15:00, while those in the Late ToD group will receive their immunotherapy after 15:00. Chemotherapy will be administered after the immunotherapy infusion, with no strict restrictions on the infusion duration.

***Reason for selecting 15:00 as the cut-off time in our clinical trial:** We performed a retrospective analysis of time-of-day effects for PFS in 447 patients receiving first line immunochemotherapy for NSCLC at our department from 2017 to 2021. We searched for the most discriminant cutoff time, using the administration of 20% of courses after cut-off, according to Qian et al. This led us to select 15:00 for our trial as supported by the attached Figure of Hazard Ratio function from this unpublished retrospective study (See figure below). Among all time points analyzed, 15:00 was associated with the lowest HR (0.36 [95% CI: 0.27–0.49], shown in red), and was therefore selected as the cut-off time for our clinical trial.



Retrospective analysis of optimal time-of-day cut-off for PFS.

Forest plots illustrating the hazard ratios (HRs) for progression-free survival (PFS), comparing patients who received immunochemotherapy before vs. after each ToDA cut-off.

All enrolled patients received 200 mg of pembrolizumab or sintilimab in 21-day cycles. For adenocarcinoma and squamous cell carcinoma, pemetrexed or nab-paclitaxel was used alongside carboplatin and immunotherapy as per guidelines. Specifically:

Squamous cell carcinoma: Patients were treated with carboplatin (AUC 5 mg/mL/min) and nab-paclitaxel (200 mg/m²) on day 1, in parallel with immunotherapy.

Adenocarcinoma: Patients received carboplatin (AUC 5 mg/mL/min) and pemetrexed (500 mg/m²) on day 1, along with immunotherapy.

Maintenance period

After 4 cycles of chemotherapy combined with pembrolizumab or sintilimab, patients will enter maintenance therapy if no disease progression occurs. For patients with histologically confirmed squamous cell carcinoma of the lung, pembrolizumab or sintilimab monotherapy will be administered in 21-day cycles, continuing until disease progression or intolerable adverse events. For patients with histologically confirmed adenocarcinoma of the lung, maintenance therapy will consist of pemetrexed in combination with pembrolizumab or sintilimab, also administered in 21-day cycles, until disease progression or intolerable adverse events. During the maintenance phase, the timing of immunotherapy sessions will not be strictly fixed.

Follow-up Period

After completing treatment, subjects will enter the follow-up period, which will primarily focus on survival. Telephone follow-ups will be conducted every 3 months to gather information on the subjects' survival status, any anti-tumor treatments received after the study, time to disease progression, and other relevant data. This follow-up will continue until the subject's death, loss to follow-up, withdrawal of informed consent, or termination of the study by the sponsor.

Tumor Imaging Assessment

Tumor assessments will be conducted periodically based on RECIST v1.1 criteria. All subjects

will undergo initial tumor imaging assessments during screening using PET-CT and enhanced head MRI. Once enrolled in the study, patients will be evaluated after every two treatment cycles, with imaging including enhanced CT scans of the chest and abdomen, as well as head MRI. During the maintenance therapy phase, tumor assessments will follow the same schedule as during the chemotherapy and immunotherapy combination phase, continuing until the end of maintenance therapy.

During the follow-up period, tumor assessments will not be emphasized; only the patients' survival status will be recorded. However, the frequency of assessments may be appropriately increased in the event of adverse events or possible disease progression.

3.2 Basis for Study Design

3.2.1 Rationale for Selection of Subject Population

Stage IIIC-IV non-small cell lung cancer (NSCLC) represents a highly heterogeneous group of diseases, typically treatable only in advanced stages or in conjunction with local therapies. The vast majority of patients are not candidates for surgery. Patients with driver mutations, such as *EGFR* mutations, or *ALK* and *ROS1* rearrangements, are often treated with targeted therapies, as they generally do not respond well to immunotherapy. Therefore, this study will exclusively include patients who are negative for driver mutations and are receiving first-line chemotherapy combined with immunotherapy.

Other inclusion and exclusion criteria will follow standard eligibility guidelines for immunotherapy to exclude subjects at high risk for adverse reactions to immunotherapy or chemotherapy. This approach will ensure the protection of participants' rights and safety, while minimizing interference in the evaluation of the study drug, subject safety, or study outcomes.

3.2.2 Rationale for Endpoint Selection

The primary objective of this study is to evaluate the efficacy of early versus late immunotherapy infusion in patients with Stage IIIC-IV non-small cell lung cancer, with progression-free survival (PFS) as the primary endpoint. Secondary endpoints include overall survival (OS) and objective response rate (ORR)

3.2.3 Randomization

Subjects who meet the study entry criteria will be randomly assigned in a 1:1 ratio to either the Early ToD group or the Late ToD group. Randomization was performed in a 1:1 ratio using a computer-generated random number table prepared by an independent statistician. Group assignments were concealed in sequentially numbered, opaque, sealed envelopes, and were not accessible to study investigators until the point of allocation.

3.3 End of Study

The end of study is defined as the date of last patient last follow-up that is used to assess the primary efficacy endpoint (PFS), or the date when the sponsor decides to end the study, whichever is earlier.

3.4 Blinded Independent Review Committee (BIRC)

The BIRC will be composed of board-certified physicians with expertise in solid tumor oncology and radiology. The committee will independently evaluate imaging data and clinical findings to assess treatment response and disease progression. BIRC assessments will include review of target and non-target lesions, as well as new lesion identification, based on pre-

defined radiologic criteria. The BIRC will conduct assessments in a blinded manner, without knowledge of the subjects' treatment groups.

All imaging will be reviewed centrally through a standardized platform to ensure consistency. Details on BIRC processes, timelines, and adjudication procedures will be described in a separate BIRC charter.

4 SELECTION AND WITHDRAWAL OF SUBJECTS

4.1 Inclusion Criteria

Patients must meet all of the following inclusion criteria to be eligible for entry into the study.

1. Understanding the requirements and content of the clinical trial, and providing a signed and dated informed consent form.
2. Male or female patients aged ≥ 18 years at the time of signing the informed consent form.
3. An Eastern Cooperative Oncology Group (ECOG) performance status of 0-1 (Appendix I).
4. Histologically or cytologically confirmed stage IIIC-IV non-small cell lung cancer (NSCLC) with no prior systemic treatment. Disease staging should be based on the 8th edition of the AJCC/UICC NSCLC staging system.
5. Presence of measurable target lesions as defined by RECIST 1.1, confirmed by CT or MRI within 28 days prior to the first dose of study drug.
6. Negative for EGFR, ALK, and ROS1 mutations, as confirmed by a certified laboratory.
7. Patients with stable brain metastases are eligible if no local treatment or steroid tapering is required.
8. FEV1 > 1.0 L and FEV1% $> 40\%$ within 3 months prior to enrollment.
9. Adequate hematologic and major organ function defined by the following laboratory criteria, with tests completed within 14 days prior to the first dose of study treatment:
 - (1) Hematology (no hematopoietic stimulating factor drugs or blood transfusion within 14 days before the first study treatment): absolute neutrophil count (ANC) $\geq 1.5 \times 10^9/\text{L}$, absolute lymphocyte count (LC) $\geq 0.5 \times 10^9/\text{L}$; platelet count (PLT) $\geq 100 \times 10^9/\text{L}$, hemoglobin (Hb) ≥ 90 g/L
 - (2) Liver function: aspartate transferase (AST) and alanine aminotransferase (ALT) $\leq 3 \times \text{ULN}$; serum total bilirubin (TBIL) $\leq 1.5 \times \text{ULN}$ (total bilirubin ≤ 3.0 mg/dL in patients with confirmed Gilbert syndrome); albumin (ALB) ≥ 3 g/dL;
 - (3) Renal function: creatinine clearance rate (CrCl) ≥ 45 mL/minute (by Cockcroft-Gault formula);
 - (4) Coagulation: international normalized ratio (INR) ≤ 1.5 , activated partial thromboplastin time (APTT) $\leq 1.5 \times \text{ULN}$;
 - (5) Cardiac color ultrasound: left ventricular ejection fraction (LVEF) $\geq 50\%$

10. Female or male patients of childbearing potential who are not surgically sterilized must agree to abstain from sexual intercourse (avoid heterosexual intercourse) or use at least one medically acceptable contraceptive (e.g., bilateral tubal ligation, male sterilization, hormonal contraceptives that inhibit ovulation, intrauterine device that releases hormones and copper-containing intrauterine device) during the study treatment period and for 180 days after the end of the study treatment period; female patients of childbearing potential who are not surgically sterilized must have a negative serum HCG test within 14 days before the first dose.

4.2 Exclusion Criteria

Patients with any of the following characteristics or conditions will not be eligible for the study:

Tumor-Related and Anti-Tumor Treatment Exclusions:

1. Patients with histologically or cytologically confirmed small cell lung cancer (SCLC), mixed SCLC and NSCLC, large-cell neuroendocrine carcinoma, or sarcomatoid carcinoma.
2. Patients with *EGFR* mutations or *ALK/ROS1* gene rearrangements. All patients must undergo *EGFR* gene testing and *ALK/ROS1* gene and/or immunohistochemistry testing.
3. Patients who have previously received systemic anti-tumor therapy for NSCLC, including investigational drugs. Additionally, those who have received traditional Chinese medicine for anti-tumor purposes must ensure a washout period of at least 2 weeks prior to the first dose of the study drug.
4. Patients who have previously received radiotherapy to the chest.
5. Patients who have participated in other clinical trials within 4 weeks or within 5 drug half-lives (whichever is shorter) prior to the first dose of the study drug. This includes those who received systemic immunostimulatory therapy (e.g., interferon or interleukin-2, including investigational immunostimulatory agents) within 4 weeks prior to the first dose.
6. Patients who received systemic immunosuppressive therapy (e.g., corticosteroids, cyclophosphamide, azathioprine, methotrexate, thalidomide, anti-tumor necrosis factor drugs) within 2 weeks prior to the first dose or who are expected to require such therapy during the study. Exceptions may be made for patients who received short-term, low-dose systemic immunosuppressive drugs (≤ 10 mg/day prednisone or equivalent) or short-term, high-dose systemic corticosteroids (e.g., for contrast agent allergies, 48-hour corticosteroid treatment), with the approval of the medical monitor. Patients with chronic obstructive pulmonary disease (COPD) on ≤ 10 mg/day prednisone or equivalent, patients with orthostatic hypotension on mineralocorticoid therapy (e.g., fludrocortisone), and patients with adrenal insufficiency on physiological replacement doses of corticosteroids may be enrolled.
7. Patients with symptomatic brain metastases requiring dehydration therapy or local radiotherapy.

Comorbidities/medical history

8. Presence of an autoimmune disorder. The following are allowed: Type 1 diabetes (blood glucose can be controlled by insulin therapy) and hypothyroidism caused by

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- autoimmune thyroiditis for which only hormone replacement therapy is indicated;
vitiligo skin disease manifestations only (psoriatic arthritis should be excluded);
9. Presence of a malignant tumor other than non-small cell lung cancer within 5 years before screening, with the exception of adequately treated cervical carcinoma in situ, basal cell or squamous epithelial cell skin cancer, local prostate cancer after radical surgery, and ductal carcinoma in situ after radical surgery (hormone treatment of non-metastatic prostate cancer or breast cancer is allowed);
 10. Patients with known or suspected interstitial pneumonitis; other moderate to severe lung diseases that may interfere with the detection or treatment of drug-related pulmonary toxicity and that seriously affect respiratory function, including, for example, idiopathic pulmonary fibrosis, organizing pneumonia/bronchiolitis obliterans, etc.;
 11. Patients with severe cardiovascular and cerebrovascular diseases, such as conditions meeting NYHA criteria (Class II or higher), or myocardial infarction or cerebrovascular accidents (cerebral hemorrhage, symptomatic cerebral infarction, etc.) within 3 months before the first dose, or unstable arrhythmia or unstable angina within 1 month before the start of study treatment;
 12. Clinically significant bleeding symptoms or a significant bleeding tendency within one month before the first dose, e.g., gastrointestinal bleeding, gastric ulcer bleeding, etc.;
 13. Arterial/venous thrombosis event within 3 months before the first dose, such as deep vein thrombosis and pulmonary embolism;
 14. Positive HIV test;
 15. Patients with active hepatitis B (defined as positive hepatitis B virus surface antigen [HBsAg] during the screening period along with HBV DNA above the upper limit of normal of the site's laboratory) or hepatitis C (defined as positive hepatitis C virus antibody [HCV-Ab] and positive HCV-RNA during the screening period);
 16. Evidence of active tuberculosis infection within 1 year before the first dose;
 17. Serious infection within 4 weeks before the first dose, including but not limited to infection complications, bacteremia and severe pneumonia requiring hospitalization or ≥ 2 weeks of intravenous antibiotic administration; active infection that has received therapeutic intravenous or oral antibiotics within 2 weeks before the start of study treatment. Patients receiving prophylactic antibiotic therapy (e.g., to prevent urinary tract infection) may be enrolled
 18. History of live attenuated vaccination within 28 days before the first dose or patients expecting live attenuated vaccination during the study;
 19. Major surgery other than diagnosis or biopsy within 28 days before the first dose; minor traumatic surgery (biopsy, bronchoscopy and chest drainage) within 7 days before the first dose;
 20. Patients who have received or are preparing to receive an allogeneic bone marrow transplantation or solid organ transplantation;
 21. History of severe allergic reactions to other monoclonal antibodies/fusion protein drugs;
 22. Hypersensitivity to any component of the randomly assigned treatment regimen;
 23. Female patients who are pregnant, breastfeeding, or planning to become pregnant during the study;

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24. Patients with a known history of psychotropic substance abuse, alcohol or drug use; and any other medical (e.g., pulmonary, metabolic, endocrine or neurological disease, congenital disease, etc.), psychiatric, or social condition that, in the opinion of the investigator, may interfere with the patient's rights, safety, health, or ability to sign the informed consent form or collaborate and participate in the study, or interfere with the evaluation of the study drug, or the interpretation of patient safety or study results.

4.3 Enrollment Criteria

Subjects who do not meet the entry criteria will not be randomized into the study. If, after enrollment but before the initiation of study treatment, it is discovered that a subject was incorrectly enrolled, incorrectly started treatment, or no longer meets the entry criteria, they should be withdrawn from the study. The study allows for re-screening of subjects who previously failed the screening process. Re-screening does not need to be repeated for tests performed within the protocol-defined time window (e.g., tumor imaging or hepatitis virus testing from the initial screening). However, any prior tests that exceed the protocol's time window must be repeated.

4.4 Subject Withdrawal from or Discontinuation of Treatment and Site Termination

Subjects must withdraw from the study in the following cases:

- The subject or their legal representative withdraws informed consent to participate in the study.
- The subject is lost to follow-up.
- Patients do not receive ICIs at the scheduled times during the initial cycles of treatment.

Subjects must discontinue the study drug but continue follow-up for the following reasons:

- Intolerable adverse reactions related to the study drug.
- Tumor progression confirmed by imaging or clinical symptoms suggestive of progression.
- General deterioration of health status, making continued participation in the study unfeasible.
- Occurrence of pregnancy in female subjects during the study.
- The subject initiates other systemic anti-tumor treatments.

The investigator determines that the subject is unable to continue treatment for other reasons, such as mental illness, social conditions that might interfere with the subject's rights, safety, cooperation, and participation in the study, or poor compliance.

5 INVESTIGATIONAL PRODUCT

Before and after chemotherapy, antiemetic therapy should be given according to clinical practice and local treatment principles. Prophylactic antiemetic therapy with corticosteroids should be avoided due to their immunomodulatory effects.

Nab-paclitaxel

Nab-paclitaxel 200 mg/m² is administered on Day 1 of each 3-week dosing cycle during the combination treatment period for a total of 4 cycles.

Allergic reactions with nab-paclitaxel injection should be prevented according to the package insert or the hospital's dosing practice. Because hormones may affect the proliferation of T

cells and thus affect the efficacy of immunotherapy, the investigator is allowed to adjust the dose of dexamethasone as appropriate, depending on the subjects' specific conditions.

Pemetrexed

Pemetrexed 500 mg/m² is administered on Day 1 of each 3-week dosing cycle during the combination treatment period and maintenance period.

Allergic reactions with pemetrexed injection should be prevented according to the package insert or the hospital's dosing practice. Because hormones may affect the proliferation of T cells and thus affect the efficacy of immunotherapy, the investigator is allowed to adjust the dose of dexamethasone as appropriate, depending on the subjects' specific conditions.

Carboplatin

AUC 5 mg/mL/min (administered according to the Calvert formula) is administered via intravenous drip on Day 1 of each 3-week dosing cycle during the combination treatment period

Pembrolizumab/Sintilimab

Pembrolizumab/Sintilimab was administered at a fixed dose of 200 mg every cycle, including the combination chemotherapy treatment period as well as the maintenance treatment period, until disease progression or serious adverse events occurred.

6 Continued treatment after Disease Progression

In this study, no special requirements or arrangements will be made for the treatment regimen of patients after disease progression or after withdrawal due to serious adverse events.

7 STUDY PROCEDURES

7.1 Screening Period

First, it should be confirmed that the subject has signed the informed consent form (ICF).

Except existing tumor imaging examinations, pathological diagnoses and genetic testing within the time limit prescribed before the first dose, a written informed consent form must be obtained in advance before any clinical study procedures are performed.

Re-screening is allowed in this study for subjects who have previously failed screening. At re-screening, they must sign the informed consent form again and re-register to obtain a new subject number.

The first dose must be no more than 28 days from the signing of the informed consent form by the subject. Subjects will be evaluated for final eligibility for study enrollment within the defined time period given in the Schedule of Activities.

Refer to the Schedule of Activities for details on the test items and procedures during the screening period.

7.2 Treatment Period

Study treatment will commence within 3 business days of enrollment. If laboratory tests conducted during screening fall within the specified time window (as outlined in the Schedule of Activities), they do not need to be repeated before the first dose. A window of ± 3 days is allowed for each cycle visit during the study. Protocol-specified study tests must be completed before each dosing cycle, and dosing may not proceed until safety assessments are completed. Instructions should be provided in the event of special circumstances.

Tumor assessments will be performed periodically using RECIST v1.1 criteria. If disease

progression is suspected (e.g., worsening clinical symptoms), and no assessment has been performed within the past 4 weeks, an additional assessment should be conducted at the end of treatment. The frequency of assessments may be increased according to the investigator's plan if clinically indicated. Following the first radiographic assessment of disease progression, a confirmatory assessment should be conducted 6 weeks later if the subject is clinically stable; if the subject is clinically unstable, no confirmation is required.

Details of specific tests and procedures required at each visit are provided in the Schedule of Activities. Adverse event evaluations and records of concomitant medications/therapies will be documented at each visit.

7.3 Survival Follow-up

Survival follow-up is a telephone follow-up every 90 days (± 7 days) from the last dose or surgery/radiotherapy to collect survival information (date of death and cause of death) and information after the end of study treatment (including subsequent anti-tumor treatment and efficacy) until the death endpoint or until the subject is lost to follow-up or the study is terminated. At certain time points (including but not limited to the phased analysis and the final analysis), the sponsor can conduct unified survival follow-up for all subjects who are still in the study.

8 Evaluation

The tests and procedures required by the protocol will be ensured as much as possible in accordance with the protocol. However, unplanned circumstances may occur from time to time, which is beyond the investigator's control, making testing difficult. Under such circumstances, the investigator must take all necessary measures to protect the safety and interests of the subjects. When a test required by the protocol cannot be performed, the investigator will record the reason. In addition, the study team must be notified of unexpected situations in a timely manner.

8.1 Efficacy Assessment Criteria

Tumor ORR, and PFS and other efficacy indicators based on the objective tumor response, will be evaluated per RECIST v1.1 using radiographic imaging (see Appendix II for details). During the study period, subjects' survival status and assessments will be continuously recorded. Subjects will be followed up by telephone on survival every 3 months after termination of treatment until death, loss to follow-up, withdrawal of informed consent, or termination of the study. The documentation and radiographic data of all subjects must be subject to source verification and peer review.

8.2 Imaging Examination

Tumor assessments will be conducted periodically based on RECIST v1.1 criteria. All subjects will undergo initial tumor imaging assessments during screening using PET-CT (preferred) or enhanced neck/chest/abdomen CT and enhanced head MRI. Once enrolled in the study, patients will be evaluated after every two treatment cycles, with imaging including enhanced CT scans of the chest and abdomen, as well as head MRI. During the maintenance therapy phase, tumor assessments will follow the same schedule as during the chemotherapy and immunotherapy combination phase, continuing until the end of maintenance therapy. During the follow-up period, tumor assessments will not be emphasized; only the patients' survival status will be recorded. However, the frequency of assessments may be appropriately

increased in the event of adverse events or possible disease progression.

8.3 Others

Vital signs, including body temperature, blood pressure, pulse, and respiratory rate, will be measured within 30 minutes before and 30-60 minutes after the completion of all drug infusions. The frequency of these tests may be increased at any time if the subject experiences any symptoms of discomfort. Resting oxygen saturation will be measured by pulse oximetry within 3 days prior to each dose and whenever the subject develops new or worsening respiratory symptoms. Oxygen levels may also be monitored prior to dosing and when new or worsening respiratory symptoms occur, as appropriate.

A complete physical examination (covering general condition, head and face, eyes, ears, nose, throat, oral cavity, skin, lymph nodes, respiratory system, cardiovascular system, gastrointestinal system, genitourinary system, musculoskeletal system, nervous system, psychiatric status, and others) will be conducted within 3 days prior to the first dose and at study treatment withdrawal. Abnormal findings noted at baseline will be recorded in the eCRF as part of the general medical history and baseline conditions. Targeted physical examinations will be performed prior to dosing on Day 1 of each subsequent cycle (the first dose examination does not need to be repeated if completed within 3 days during the Screening Period). Physical examinations, weight measurements, and ECOG performance scores will be assessed on each dosing day. Clinically significant new or worsening abnormalities will be documented at subsequent visits in the Adverse Events section of the eCRF.

9 ADVERSE EVENT REPORTING Adverse Event (AE)

The collection of AE information starts from the signing of the informed consent form by the subject until the end of the safety follow-up period (90 ± 7 days after the last dose of immunotherapy, whichever occurs last).

9.1 Definition of Adverse Event

Adverse events (AEs) refer to all untoward medical events that occur after a subject receives the investigational product, which may manifest as symptoms, signs, diseases or laboratory abnormalities, but do not necessarily have a causal relationship with the investigational product. At least the followings are included:

- 1) Exacerbation of a pre-existing (before entering the clinical trial) medical condition/disease (including exacerbation of symptoms, signs, and laboratory abnormalities);
- 2) Any new AEs: Any new adverse medical conditions (including symptoms, signs and newly diagnosed diseases);
- 3) Clinically significant abnormal laboratory test values or results that are not caused by comorbidities.

The study staff should record in detail any AE experienced by each subject in the AE page of the eCRF, including but not limited to: the name of the AE and description of all associated symptoms, time of onset, severity, relationship with the study drug, duration, measures taken for the study drug, and the final outcome.

9.2 Criteria for Determination of Severity of Adverse Events

Refer to NCI-CTCAE v5.0 for grading criteria for adverse drug reactions. Refer to the criteria in Table 1 if AEs not listed in the table of NCI-CTCAE v5.0 occur:

**Table 1. Reference Criteria for Determination of Severity of AEs Not Listed in Table of NCI-CTC AE
Version 5.0**

Grade	Clinical description of severity
1	Mild; asymptomatic or with mild clinical symptoms; clinical or laboratory abnormalities only; intervention not indicated.
2	Moderate; minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental activities of daily living, which refer to preparing meals, shopping, using the telephone, counting money, etc.
3	Severe or medically significant symptoms but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care activities of daily living. Self-care activities of daily living refer to: bathing, dressing and undressing, feeding self, using the toilet, taking medications, and ambulatory.
4	Life-threatening, urgent intervention indicated.
5	Death due to adverse event

9.3 Immune-related Adverse Events

Immune-related adverse events are specific events (including pneumonia, diarrhea/colitis, hepatitis, nephritis/renal dysfunction, rash, and endocrinopathy) when the subject needs to receive immunosuppressant drugs. Exceptions are usually endocrinopathy events (hypothyroidism/thyroiditis, hyperthyroidism, hypophysitis, diabetes mellitus, adrenal insufficiency), which, although immune-related, are usually not related to therapy and can be managed without immunosuppressive interventions. Preferred terms included in the IMAE analysis to support warnings and precautions are shown in the table below (Table 2).

Table 2 Preferred Terms Included in IMAE Analysis to Support Warnings and Precautions

IMAE category	PT terms Included in IMAE category (MedDRA coding)
Pneumonia	Pneumonia, interstitial pneumonitis
Diarrhea/colitis	Diarrhea, colitis, enterocolitis
Hepatitis	Hepatotoxicity, hepatitis, hepatitis acute, autoimmune hepatitis, AST increased, ALT increased, bilirubin increased, ALP increased
Adrenal insufficiency	Adrenal insufficiency
Hypothyroidism/thyroiditis	Acute thyroiditis (prostration with frequent thyroiditis), autoimmune thyroiditis (prostration with frequent thyroiditis)
Hyperthyroidism	Hyperthyroidism
Hypophysitis	Hypophysitis
Diabetes	Diabetes mellitus, diabetic ketoacidosis
Nephritis and renal dysfunction	Nephritis, acute nephritis, tubulointerstitial nephritis, renal failure acute, renal failure, creatinine increased
Rash	Rash, rash maculo-papular

10 DATA ANALYSIS/STATISTICAL METHODS

10.1 Analysis Set

Intent-to-Test (ITT) Population: This population includes all randomized patients, and analyses are performed based on their originally assigned treatment groups, regardless of whether they completed the intervention as planned. This dataset serves as the primary

population for evaluating efficacy outcomes.

The Full Analysis Set (FAS): FAS included all randomized patients who received at least one dose of the study treatment as planned.

Per-protocol Set (PPS): includes patients who strictly adhere to the treatment regimen, excluding subjects with major protocol deviations that are judged to have a significant impact on the results.

10.2 Sample Size

The primary endpoint of this study is the Progression-Free Survival (PFS) of patients receiving treatment. The sample size is determined based on the PFS of the two groups.

The study design includes two cohorts: the Early ToD group, where the first four immunotherapy infusions are administered before 15:00, and the Late ToD group, where the first four immunotherapy infusions are administered after 15:00. Based on data from previous retrospective studies, it is assumed that the median PFS for the Late ToD group is 6 months, while the median PFS for the Early ToD group is 10 months. The two groups will be randomized in a 1:1 ratio. Statistical assumptions include a two-sided $\alpha = 0.05$ and 80% power. The enrollment period is assumed to be 12 months, with a 24-month follow-up period and a dropout rate of 5%. The final calculated total sample size for the two groups is 210 patients including 105 in Early ToD group and 105 in Late ToD group.

10.3 Statistical Analysis Method

10.3.1 General Analysis

The data of this study will be summarized in descriptive statistics according to the following general principles, unless otherwise specified.

Categorical data will be summarized descriptively using statistics including frequency and percentage, with the 95% confidence interval provided for the overall percentage when necessary. Continuous data will be summarized descriptively using statistics including mean, standard deviation (SD), standard error of the mean (SEM), median, minimum and maximum. Time-to-event data will be summarized using the Kaplan-Meier method to estimate survival function, plot survival curves, and estimate median time and its 95% confidence interval.

In the statistical analysis, group comparisons will be conducted between the Early and Late ToD groups, unless otherwise specified. Subgroup analyses will also be performed based on the baseline characteristics of the patients throughout the study.

10.3.2 Efficacy Analysis

10.3.2.1 Primary Efficacy Endpoint

The primary efficacy endpoints of this study are the PFS as assessed by the BIRC after ICIs combination with chemotherapy.

PFS

For the analysis of the primary efficacy endpoint, survival curves will be plotted using the Kaplan-Meier method and the median survival time will be estimated for each cohort. Standard errors will be calculated using Greenwood's method and 2-sided 95% confidence intervals for median survival time will be estimated for each cohort using the log-log transform method.

10.3.2.2 Secondary efficacy endpoints

The Secondary efficacy endpoints of this study are the OS and ORR as assessed by the BIRC after ICIs combination with chemotherapy.

OS

For the analysis of the OS, survival curves will be plotted using the Kaplan-Meier method and the median survival time will be estimated for each cohort. Standard errors will be calculated using Greenwood's method and 2-sided 95% confidence intervals for median survival time will be estimated for each cohort using the log-log transform method.

ORR

Descriptive statistics will be used to summarize the ORR of each cohort. Clopper Pearson and Wilson methods will also be used to estimate the 95% confidence interval for the ORR of each cohort. For subjects who do not complete the ORR assessment, the imputation method will be described in the statistical analysis plan.

11 DATA MANAGEMENT METHODS

11.1 Data Collection

11.1.1 Completion of Electronic Case Report Form (eCRF)

The eCRFs will be completed by the investigator or data entry personnel. The eCRFs should be completed in a timely manner and the completed data must be traceable to the original records. When modifying data, the reason for data modification shall be filled in according to the system prompts. The investigator or authorized designee must confirm the authenticity, completeness, and timeliness of the eCRF data and provide an electronic signature.

11.1.2 Review of eCRF Data

The investigator or data entry personnel can make corrections to the data or explain and confirm the data by responding to the queries. The monitor, data administrator and medical reviewer will also review the eCRF data and issue queries about questionable data when necessary. The investigator should promptly respond to queries from the system and data reviewers. When necessary, multiple queries may be issued until the data issue is resolved.

12 QUALITY ASSURANCE AND QUALITY CONTROL

In order to ensure the quality of the study, the sponsor and investigator will jointly discuss and develop the clinical study protocol before the study officially starts. They will also confirm whether the study personnel participating in the study have undergone appropriate GCP training.

According to the GCP guidelines, necessary procedures should be taken during the design and implementation stages of the study to ensure that the collected data are accurate, consistent, complete and credible. All observed results and abnormal findings in the clinical trial shall be verified and recorded promptly to ensure the reliability of the data. The instruments, equipment, reagents and standards, among others, used for all examinations/tests in the clinical trial shall be subject to strict quality standards, and their use under normal conditions shall be ensured.

The investigator will enter the information required by the protocol into the eCRF. The monitor will verify whether the information is complete and accurate, and guide the staff of the study site to make necessary corrections and supplementations.

The drug regulatory authority, the Ethics Committee, the sponsor's monitors and/or auditors may conduct systematic inspections on relevant activities and documents of the clinical trial to evaluate whether the study is conducted in accordance with the requirements of the study

protocol, SOPs and relevant regulations, and whether the study data are recorded in a timely, truthful, accurate and complete manner. The audit shall be conducted by people not directly involved in the clinical trial.

13 REGULATORY ETHICS, INFORMED CONSENT AND SUBJECT PROTECTION

13.1 Institutional Review Board/Independent Ethics Committee

Prior to initiation of the study, the investigator must obtain dated written approval/consent from the Institutional Review Board/Independent Ethics Committee (IRB/IEC) for the protocol, informed consent form, subject recruitment materials (e.g., advertisements), and all other written materials to be provided to the subjects.

During the clinical study, any amendments made to this protocol should be submitted to the IRB/IEC. If necessary, other study documents should also be amended accordingly, and submitted and/or approved in accordance with the requirements of the IRB/IEC.

It is the responsibility of the investigator to submit the study interim report regularly in accordance with the relevant requirements of the IRB/IEC, and inform the IRB/IEC of the study completion after the end of the study.

13.2 Confidentiality of Subject Information

The subject information will be kept strictly confidential by the investigator, participating study staff, sponsor and its agents. Confidentiality covers biological samples and genetic tests in addition to subjects' clinical information. Therefore, the protocol, documentations, data and all other information generated therefrom will be kept strictly confidential. Without the prior written approval of the sponsor, all relevant study or data information shall not be disclosed to any unauthorized third party.

The study monitor, other authorized representatives of the sponsor, IRB, or other representatives authorized by the sponsor may check all documents and records that the investigator is required to maintain, including but not limited to medical records and medication records of subjects. The study site should allow access to these records.

The contact information of subjects will be securely stored at each study site and will only be used internally during the study process. At the end of the study, all records will continue to be stored in a secure place in accordance with the time limit specified by the local IRB and regulations.

Appendix I Performance Status Scale (ECOG)

(Eastern Cooperative Oncology Group)

Grade	Description
0	Asymptomatic, fully active, able to carry on all pre-disease performance without restriction.
1	Symptomatic, restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work.
2	Symptomatic, ambulatory and capable of all selfcare but unable to carry out any work activities; up and about more than 50% of waking hours.
3	Symptomatic, capable of only limited selfcare; confined to bed or chair more than 50% of waking hours.
4	Completely disabled; cannot carry on any selfcare; totally confined to bed or chair.
5	Dead

Oken, M.M., Creech, R.H., Tormey, D.C., Horton, J., Davis, T.E., McFadden, E.T.,

Carbone, P.P.: Toxicity And Response Criteria Of The Eastern Cooperative

Oncology Group. Am J Clin Oncol 5: 649 - 655, 1982

Appendix II Response Evaluation Criteria in Solid Tumors (RECIST v1.1)

Response Evaluation Criteria in Solid Tumors Version 1.1 (Excerpt)

1 Background

Omitted

1 Purpose

Omitted

3 Measurability of Tumor at Baseline

3.1 Definitions

At baseline, tumor lesions/lymph nodules will be categorized measurable or non-measurable as follows:

3.1.1 Measurable

Tumor lesions: Must be accurately measured in at least one dimension (longest diameter in the plane of measurement is to be recorded) with a minimum size of:

- 10 mm by CT scan (CT scan slice thickness no greater than 5 mm);
- 10 mm caliper measurement by clinical exam (lesions which cannot be accurately measured with calipers should be recorded as non-measurable);
- 20 mm by chest X-ray;
- Malignant lymph nodes: To be considered pathologically enlarged and measurable, a lymph node must be ≥ 15 mm in short axis when assessed by CT scan (CT scan slice thickness recommended to be no greater than 5 mm). At baseline and in follow-up, only the short axis will be measured and followed.

3.1.2 non-measurable

All other lesions, including small lesions (longest diameter <10 mm or pathological lymph nodes with ≥ 10 mm to <15 mm short axis) as well as truly non-measurable lesions. Lesions considered truly non-measurable include: leptomeningeal disease, ascites, pleural or pericardial effusion, inflammatory breast disease, lymphangitic involvement of skin or lung, abdominal masses/abdominal organomegaly identified by physical exam that is not measurable by reproducible imaging techniques, and cystic lesions.

3.1.3 Special considerations regarding lesion measurability

Bone lesions, cystic lesions, and lesions previously treated with local therapy require

particular comment:

Bone lesions:

- Bone scan, PET scan or plain films are not considered adequate imaging techniques to measure bone lesions. However, these techniques can be used to confirm the presence or disappearance of bone lesions;
- Lytic lesions or mixed lytic-blastic lesions, with identifiable soft tissue components, that can be evaluated by cross-sectional imaging techniques such as CT or MRI can be considered as measurable lesions if the soft tissue component meets the definition of measurability described above;
- Blastic bone lesions are non-measurable lesions.

Cystic lesions:

- Lesions that meet the criteria for radiographically defined simple cysts should not be considered as malignant lesions (neither measurable nor non-measurable) since they are, by definition, simple cysts.
- Cystic lesions thought to represent cystic metastases can be considered as measurable lesions, if they meet the definition of measurability described above. However, if non-cystic lesions are present in the same patient, these are preferred for selection as target lesions.

Lesions with prior local treatment:

- Tumor lesions situated in a previously irradiated area, or in an area subjected to other loco-regional therapy, are usually not considered measurable unless there has been demonstrated progression in the lesion. Study protocols should detail the conditions under which such lesions would be considered measurable.

3.2 Specifications by methods of measurements

3.2.1 Measurement of lesions

All measurements should be recorded in metric notation during clinical evaluation. All baseline evaluations should be performed as close as possible to the treatment start and never more than 28 days (4 weeks) before the beginning of the treatment.

3.2.2 Method of assessment

The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging based evaluation

should always be done rather than clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical exam.

Clinical lesions: Clinical lesions will only be considered measurable when they are superficial and ≥ 10 mm diameter as assessed using calipers (e.g., skin nodules). For the case of skin lesions, documentation by color photography including a ruler to estimate the size of the lesion is suggested. As noted above, when lesions can be evaluated by both clinical exam and imaging, imaging evaluation should be undertaken since it is more objective and may also be reviewed at the end of the study.

Chest X-ray: Chest CT is preferred over chest X-ray, especially when tumor progression is an important clinical endpoint, since CT is more sensitive than X-ray, particularly in identifying new lesions. However, lesions on chest X-ray may be considered measurable if they display clear borders and are surrounded by aerated lung.

CT, MRI: CT is the best currently available and reproducible method to measure lesions selected for response assessment. This guideline has defined measurability of lesions on CT scan based on the assumption that CT slice thickness is 5 mm or less. When CT scans have slice thickness greater than 5 mm, the minimum size for a measurable lesion should be twice the slice thickness. MRI is also acceptable in certain situations (e.g. for body scans).

Ultrasound: Ultrasound is not useful in assessment of lesion size and should not be used as a method of measurement. Ultrasound examinations cannot be reproduced in their entirety for independent review at a later date and, because they are operator dependent, it cannot be guaranteed that the same technique and measurements will be taken from one assessment to the next. If new lesions are identified by ultrasound in the course of the study, confirmation by CT or MRI is advised. If there is concern about radiation exposure at CT, MRI can be used instead of CT in selected instances.

Endoscopy, laparoscopy: The utilization of these techniques for objective tumor evaluation is not advised. However, they can be useful to confirm complete pathological response when biopsies are obtained or to determine relapse in trials where recurrence following complete pathological response or surgical resection is an endpoint.

Tumor markers: Tumor markers alone cannot be used to assess objective tumor response. If markers are initially above the upper normal limit, however, they must normalize for a patient

to be considered in complete response. Because tumor markers are disease specific, instructions for their measurement should be incorporated into protocols on a disease specific basis. Specific criteria for both CA-125 response (in recurrent ovarian cancer) and PSA response (in recurrent prostate cancer) have been published. In addition, the Gynecologic Cancer Intergroup has developed CA-125 progression criteria which are to be integrated with objective tumor assessment for use in first-line trials in ovarian cancer.

Cytology/histology: These techniques can be used to differentiate between PR and CR if required by protocol (for example, residual lesions in tumor types such as germ cell tumors, where known residual benign tumors may remain). When effusions are known to be a potential adverse effect of treatment (e.g., with certain taxane compounds or angiogenesis inhibitors), the cytological confirmation of the neoplastic origin of any effusion that appears or worsens during treatment can be considered if the measurable tumor has met the criteria for response or stable disease in order to differentiate between response (or stable disease) and progressive disease.

4 Tumor response evaluation

4.1 Assessment of overall tumor burden and measurable disease

To assess objective response or future progression, it is necessary to estimate the overall tumor burden at baseline and use this as a comparator for subsequent measurements. Only patients with measurable disease at baseline should be included in protocols where objective tumor response is the primary endpoint. Measurable disease is defined by the presence of at least one measurable lesion. In studies where the primary endpoint is tumor progression (either time to progression or proportion with progression at a fixed date), the protocol must specify if entry is restricted to those with measurable disease or whether patients having non-measurable disease only are also eligible.

4.2 Baseline documentation of target and non-target lesions

When more than one measurable lesion is present at baseline all lesions up to a maximum of five lesions total (and a maximum of two lesions per organ) representative of all involved organs should be identified as target lesions and will be recorded and measured at baseline (this means in instances where patients have only one or two organ sites involved a maximum of two and four lesions respectively will be recorded).

Target lesions should be selected on the basis of their size (lesions with the longest diameter), be representative of all involved organs, but in addition should be those that lend themselves to reproducible repeated measurements. It may be the case that, on occasion, the largest lesion does not lend itself to reproducible measurement in which circumstance the next largest lesion which can be measured reproducibly should be selected.

Lymph nodes merit special mention since they are normal anatomical structures which may be visible by imaging even if not involved by tumor. Pathological nodes which are defined as measurable and may be identified as target lesions must meet the criterion of a short axis of ≥ 15 mm by CT scan. Only the short axis of these nodes will contribute to the baseline sum.

The short axis of the node is the diameter normally used by radiologists to judge if a node is involved by solid tumor. Nodal size is normally reported as two dimensions in the plane in which the image is obtained (for CT scan this is almost always the axial plane; for MRI the plane of acquisition may be axial, sagittal or coronal). The smaller of these measures is the short axis. For example, an abdominal node which is reported as being 20 mm $\times$ 30 mm has a short axis of 20 mm and qualifies as a malignant, measurable node. In this example, 20 mm should be recorded as the node measurement. All other pathological nodes (those with short axis ≥ 10 mm but < 15 mm) should be considered non-target lesions. Nodes that have a short axis < 10 mm are considered non-pathological and should not be recorded or followed.

A sum of the diameters (longest for non-nodal lesions, short axis for nodal lesions) for all target lesions will be calculated and reported as the baseline sum diameters. If lymph nodes are to be included in the sum, then as noted above, only the short axis is added into the sum. The baseline sum diameters will be used as reference to further characterize any objective tumor regression in the measurable dimension of the disease.

All other lesions (or sites of disease) including pathological lymph nodes should be identified as non-target lesions and should also be recorded at baseline. Measurements are not required and these lesions should be followed as “present”, “absent”, or in rare cases “unequivocal progression”. In addition, it is possible to record multiple nontarget lesions involving the same organ as a single item on the case record form (e.g., multiple enlarged pelvic lymph nodes or multiple liver metastases).

4.3 Response criteria

4.3.1 Evaluation of target lesions

Complete Response (CR): Disappearance of all target lesions. Any pathological lymph nodes (whether target or non-target nodes) must have reduction in short axis to < 10 mm.

Partial Response (PR): At least a 30% decrease in the sum of diameters of target lesions, taking as reference the baseline sum diameters.

Progressive Disease (PD): At least a 20% increase in the sum of diameters of target lesions, taking as reference the smallest sum on study (this includes the baseline sum if that is the smallest on study). In addition, the sum must also demonstrate an absolute increase of at least 5 mm (Note: the appearance of one or more new lesions is also considered progressive disease).

Stable Disease (SD): Neither sufficient shrinkage to qualify for PR nor sufficient increase to qualify for PD, taking as reference the smallest sum diameters while on study.

4.3.2 Special notes on the assessment of target lesions

Lymph nodes: Lymph nodes identified as target lesions should always have the actual short axis measurement recorded (measured in the same anatomical plane as the baseline examination), even if the nodes regress to below 10 mm on study. This means that when lymph nodes are included as target lesions, the ‘sum’ of lesions may not be zero even if complete response criteria are met, since a normal lymph node is defined as having a short axis of < 10 mm. Case report forms or other data collection methods may therefore be designed to have target nodal lesions recorded in a separate section where, in order to qualify for CR, each node must achieve a short axis < 10 mm. For PR, SD and PD, the actual short axis measurement of the nodes is to be included in the sum of target lesions.

Target lesions that become too small to measure: While on study, all lesions (nodal and non-nodal) recorded at baseline should have their actual measurements recorded at each subsequent evaluation, even when very small (e.g. 2 mm). However, sometimes lesions or lymph nodes which are recorded as target lesions at baseline become so faint on CT scan that the radiologist may not feel comfortable assigning an exact measure and may report them as being ‘too small to measure’. When this occurs, it is important that a value be recorded on the case report form. If it is the opinion of the radiologist that the lesion has likely disappeared, it should be recorded as 0 mm. If the lesion is believed to be present and is faintly seen, and an

accurate measurement value cannot be given, a default value of 5 mm can be assigned (Note: It is less likely that this rule will be used for lymph nodes since they usually have a definable size when normal and are frequently surrounded by fat such as in the retroperitoneum; however, if a lymph node is believed to be present and is faintly seen but too small to measure, a default value of 5 mm should be assigned in this circumstance as well). This default value of 5 mm is derived from the CT slice thickness (but this value should not be changed with varying CT slice thickness). The measurement of these lesions is potentially non-reproducible, therefore providing this default value will prevent false responses or progressions based upon measurement error. To reiterate, however, if the radiologist is able to provide an actual measure, that should be recorded, even if it is below 5 mm.

Lesions that split or coalesce on treatment: When non-nodal lesions ‘fragment’, the longest diameters of the fragmented portions should be added together to calculate the target lesion sum. Similarly, as lesions coalesce, a plane between them may be maintained that would aid in obtaining maximal diameter measurements of each individual lesion. If the lesions have truly coalesced such that they are no longer separable, the vector of the longest diameter in this instance should be the maximal longest diameter for the coalesced lesion.

4.3.3 Evaluation of non-target lesions

This section provides the definitions of the criteria used to determine the tumor response for the group of non-target lesions. While some non-target lesions may actually be measurable, they need not be measured and instead should be assessed only qualitatively at the time points specified in the protocol.

Complete Response (CR): Disappearance of all non-target lesions and normalization of tumor marker level. All lymph nodes must be non-pathological in size (< 10 mm short axis).

Non-complete response/non-progressive disease: Persistence of one or more non-target lesion(s) and/or maintenance of tumor marker level above the normal limits.

Progressive disease: Unequivocal progression of existing non-target lesions. Note:

Appearance of one or more new lesions is also considered progression.

4.3.4 Special notes on evaluation of progression of non-target disease

The concept of progression of non-target disease requires additional explanation as follows:

When the patient also has measurable disease. In this setting, to achieve ‘unequivocal

progression’ on the basis of the non-target disease, there must be an overall level of substantial worsening in non-target disease such that, even in presence of SD or PR in target disease, the overall tumor burden has increased sufficiently to merit discontinuation of therapy. A modest ‘increase’ in the size of one or more non-target lesions is usually not sufficient to qualify for unequivocal progression status. The designation of overall progression solely on the basis of change in non-target disease in the face of SD or PR of target disease will therefore be extremely rare.

When the patient has only non-measurable non-target lesions: This circumstance arises in some phase 3 trials when it is not a criterion of study entry to have measurable disease. The same general concepts apply here as noted above, however, in this instance there is no measurable disease assessment to factor into the interpretation of an increase in non-measurable disease burden. Because worsening in non-target lesions cannot be easily quantified (by definition: if all lesions are truly non-measurable), a useful test that can be applied when assessing patients for unequivocal progression is to consider if the increase in overall disease burden based on the change in non-target lesions is comparable in magnitude to the increase that would be required to declare progressive disease for target lesions: i.e., an increasing tumor burden representing an additional 73% increase in volume (which is equivalent to a 20% increase diameter in a measurable lesion) Examples include an increase in a pleural effusion from “trace” to “large”, an increase in lymphangitic disease from “localized” to “widespread”, or may be described in protocols as “sufficient to require a change in treatment”. Or an increase in a pleural effusion from trace to large, lymphatic involvement spreading from the primary site to distant sites, or may be described in protocols as “a change in treatment is necessary”. If ‘unequivocal progression’ is seen, the patient should be considered to have had overall PD at that point. While it would be ideal to have objective criteria to apply to non-measurable disease, the very nature of that disease makes it impossible to do so, therefore the increase must be substantial.

4.3.5 New lesions

The appearance of new malignant lesions denotes disease progression; therefore, some comments on detection of new lesions are very important. There are no specific criteria for the identification of new radiographic lesions; however, the finding of a new lesion should be

unequivocal: i.e., not attributable to differences in scanning technique, change in imaging modality or findings thought to represent something other than tumor (for example, some new bone lesions may be simply healing or flare of pre-existing lesions). This is particularly important when the patient's baseline lesions show partial or complete response. For example, necrosis of a liver lesion may be reported on a CT scan report as a new cystic lesion, which it is not.

A lesion identified on a follow-up study in an anatomical location that was not scanned at baseline is considered a new lesion and will indicate disease progression. An example of this is the patient who has visceral disease at baseline and while on study has a CT or MRI brain ordered which reveals metastases. The patient's brain metastases are considered to be evidence of PD even if he/she did not have brain imaging at baseline.

If a new lesion is equivocal, for example because of its small size, continued therapy and follow-up evaluation will clarify if it represents a truly new disease. If repeat scans confirm there is definitely a new lesion, then progression should be declared using the date of the initial scan.

While FDG-PET response assessments need additional study, it is sometimes reasonable to incorporate the use of FDG-PET scanning to complement CT scanning in assessment of progression (particularly possible 'new' disease). New lesions on the basis of FDG-PET imaging can be identified according to the following algorithm:

Negative FDG-PET at baseline, with a positive FDG-PET scan at follow-up is a sign of PD based on a new lesion.

No FDG-PET at baseline and a positive FDG-PET at follow-up:

If the positive FDG-PET at follow-up corresponds to a new site of disease confirmed by CT, this is PD.

If the positive FDG-PET at follow-up is not confirmed as a new site of disease on CT, additional follow-up CT scans are needed to determine if there is truly progression occurring at that site (if so, the date of PD will be the date of the initial abnormal FDG-PET scan).

If the positive FDG-PET at follow-up corresponds to a pre-existing site of disease on CT that is not progressing on the basis of the anatomic images, this is not PD.

4.4 Evaluation of best overall response

The best overall response is the best response recorded from the start of the study treatment until the end of treatment taking into account any requirement for confirmation. On occasion a response may not be documented until after the end of therapy so protocols should be clear if post-treatment assessments are to be considered in determination of best overall response. Protocols must specify how any new therapy introduced before progression will affect best response designation. The patient's best overall response assignment will depend on the findings of both target and non-target disease and will also take into consideration the appearance of new lesions. Furthermore, depending on the nature of the study and the protocol requirements, it may also require confirmatory measurement.

4.4.1 Time point response

It is assumed that at each protocol specified time point, a response assessment occurs. Table 1 provides a summary of the overall response status calculation at each time point for patients who have measurable disease at baseline.

Table 1 Time Point Response: Patients with Target (± Non-target) Lesions

Target lesions	Non-target lesions	New lesions	Overall response
CR	CR	No	CR
CR	Non-CR/non-PD	No	PR
CR	Not evaluated	No	PR
PR	Non-PD or not all evaluated	No	PR
SD	Non-PD or not all evaluated	No	SD
Not all evaluated	Non-PD	No	NE
PD	Any	Yes or No	PD
Any	PD	Yes or No	PD
Any	Any	Yes	PD

CR = complete response, **PR** = partial response, **SD** = stable disease, **PD** = progressive disease, and **NE** = in evaluable.

When patients have non-measurable (therefore non-target) disease only, Table 2 is to be used.

Table 2 Time Point Response - Patients with Non-Target Lesions Only

Non-target lesions	New lesions	Overall response
CR	No	CR

Non-target lesions	New lesions	Overall response
Non-CR or non-PD	No	Non-CR or non-PD*
Not all evaluated	No	NE
Equivocal PD	Yes or No	PD
Any	Yes	PD

*a: 'Non-CR/non-PD' is preferred over 'stable disease' for non-target disease since SD is increasingly used as an endpoint for assessment of efficacy in some trials, so to assign this category when no lesions can be measured is not advised.

4.4.2 Missing assessments and in evaluable designation

When no imaging/measurement is done at all at a particular time point, the patient is not evaluable (NE) at that time point. If only a subset of lesion measurements are made at an assessment, usually the case is also considered NE at that time point, unless a convincing argument can be made that the contribution of the individual missing lesion(s) would not change the assigned time point response. This would be most likely to happen in the case of PD. For example: For example, if a patient had a baseline sum of 50 mm with three measured lesions and at follow-up only two lesions were assessed, but those gave a sum of 80 mm, the patient will have achieved PD status, regardless of the contribution of the missing lesion.

4.4.3 Best overall response: all time points

The best overall response is determined once all the data for the patient is known. Best response determination in trials where confirmation of complete or partial response IS NOT required: Best response in these trials is defined as the best response across all time points (for example, a patient who has SD at first assessment, PR at second assessment, and PD on last assessment has a best overall response of PR). When SD is believed to be best response, it must also meet the protocol specified minimum time from baseline. If the minimum time is not met when SD is otherwise the best time point response, the patient's best response depends on the subsequent assessments. For example: For example, a patient who has SD at first assessment, PD at second and does not meet minimum duration for SD, will have a best response of PD. The same patient lost to follow-up after the first SD assessment would be considered in evaluable.

4.4.4 Special notes on response assessment

When nodal disease is included in the sum of target lesions and the nodes decrease to 'normal' size (< 10 mm), they may still have a measurement reported on scans. This measurement should be recorded even though the nodes are normal in order not to overestimate progression should it be based on increase in size of the nodes. As noted earlier, this means that patients with CR may not have a total sum of 'zero' on the case report form (CRF).

Patients with a global deterioration of health status requiring discontinuation of treatment without objective evidence of disease progression at that time should be reported as symptomatic deterioration. Every effort should be made to document objective progression even after discontinuation of treatment. Symptomatic deterioration is not a description of an objective response: it is a reason for stopping study therapy. The objective response status of such patients is to be determined by evaluation of target and non-target disease as shown in Tables 1 to 2.

Conditions that define early progression, early death and in evaluability are study specific and should be clearly described in each protocol (depending on treatment duration and treatment periodicity).

In some circumstances it may be difficult to distinguish residual disease from normal tissue. When the evaluation of complete response depends upon this determination, it is recommended that the residual lesion be investigated (fine needle aspirate/biopsy) before assigning a status of complete response. FDG-PET may be used to upgrade a response to a CR in a manner similar to a biopsy in cases where a residual radiographic abnormality is thought to represent fibrosis or scarring. The use of FDG-PET in this circumstance should be prospectively described in the protocol and supported by specialist medical literature for this situation. However, it must be acknowledged that both approaches may lead to false positive CR due to limitations of FDG- PET and biopsy resolution/sensitivity.

For equivocal findings of progression (e.g. very small and uncertain new lesions; cystic changes or necrosis in existing lesions), treatment may continue until the next scheduled assessment. If at the next scheduled assessment, progression is confirmed, the date of progression should be the earlier date when progression was suspected.

4.5 Frequency of tumor re-evaluation

Frequency of tumor re-evaluation while on treatment should be protocol specific and adapted to the type and schedule of treatment. However, in the context of phase 2 studies where the beneficial effect of therapy is not known, follow-up every 6–8 weeks (timed to coincide with the end of a cycle) is reasonable. Smaller or greater time intervals than these could be justified in specific regimens or circumstances. The protocol should specify which organ sites are to be evaluated at baseline (usually those most likely to be involved with metastatic disease for the tumor type under study) and how often evaluations are repeated. Normally, all target and non-target sites are evaluated at each assessment. In selected circumstances certain non-target organs may be evaluated less frequently. For example, bone scans may need to be repeated only when complete response is identified in target disease or when progression in bone is suspected.

After the end of the treatment, the need for repetitive tumor evaluations depends on whether the trial has as a goal the response rate or the time to an event (progression/death). If time to an event (e.g., time to progression, disease-free survival, progression-free survival) is the main endpoint of the study, then routine scheduled re-evaluation of protocol specified sites of disease is warranted. In randomized comparative trials in particular, the scheduled assessments should be performed as identified on a calendar schedule (for example: every 6–8 weeks on treatment or every 3–4 months after treatment) and should not be affected by delays in therapy, drug holidays or any other events that might lead to imbalance in a treatment arm in the timing of disease assessment.²

4.6 Confirmatory measurement/duration of response

4.6.1 Confirmation

In non-randomized trials where response is the primary endpoint, confirmation of PR and CR is required to ensure responses identified are not the result of measurement error. This will also permit appropriate interpretation of results in the context of historical data where response has traditionally required confirmation in such trials. However, in all other circumstances, i.e., in randomized trials (phase 2 or 3) or studies where stable disease or progression are the primary endpoints, confirmation of response is not required since it will not add value to the interpretation of trial results. However, elimination of the requirement for response confirmation may increase the importance of central review to protect against bias,

in particular in studies which are not blinded.

In the case of SD, measurements must have met the SD criteria at least once after study entry at a minimum interval (in general not less than 6-8 weeks) that is defined in the study protocol.

4.6.2 Duration of overall response

2 Disease free survival

The duration of overall response is measured from the time measurement criteria are first met for CR/PR (whichever is first measured) until the first date that recurrent or progressive disease is objectively documented (taking as reference for progressive disease the smallest measurements recorded on study). The duration of overall complete response is measured from the time measurement criteria are first met for CR until the first date that recurrent or progressive disease is objectively documented.

4.6.3 Duration of stable disease

Stable disease is measured from the start of the treatment (in randomized trials, from date of randomization) until the criteria for progression are met, taking as reference the smallest sum on study (if the baseline sum is the smallest, this is the reference for calculation of PD). The clinical relevance of the duration of stable disease varies in different studies and diseases. If the proportion of patients achieving stable disease for a minimum period of time is an endpoint of importance in a particular trial, the protocol should specify the minimal time interval required between two measurements for determination of stable disease.

Note: The duration of response and stable disease as well as the progression-free survival are influenced by the frequency of follow-up after baseline evaluation. It is not in the scope of this guideline to define a standard follow-up frequency. The frequency should take into account many parameters including disease types and stages, treatment periodicity and standard practice. However, these limitations of the precision of the measured endpoint should be taken into account if comparisons between trials are to be made.